INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

SBT-DF204: Computer Forensics Case Studies

Case Study 3: Investigating Suspected Illegal File Transfer From Memory Evidence

Individual Forensic Investigation Report

Field	Detail
Student Name	Mary-Joy Adewole
Registration Number	2025/FWSD/11468
Institution	International Cybersecurity and Digital Forensics Academy (ICDFA)
Course Code	SBT-DF204
Course Title	Computer Forensics Case Studies
Assessment	Case Study 3: Individual forensic investigation
Case Title	Investigating Suspected Illegal File Transfer From Memory Evidence
Evidence File	memdumpWin7.mem (Windows 7 SP1 x86 memory image), 1,073,676,288 bytes
Analysis Date	18 October 2026 (evidence copied 17:36:42 UTC); report dated 19 October 2026
Tools	Kali Linux 2026.2 (VMware Workstation); Volatility Foundation Volatility Framework 2.6.1 (python2 vol.py); sha256sum; md5sum; stat; file; strings
Time Basis	All event times are UTC as printed by Volatility. The image local time (UTC-8) is given only for reference and is never mixed into the timeline.

1. Case Reference and Student Identification

This is the individual submission of Mary-Joy Adewole (registration number 2025/FWSD/11468) for SBT-DF204 Case Study 3. The scenario is a training exercise: a memory image from a workstation is examined to decide whether browser activity, a command prompt, a removable device and a user profile form a defensible sequence for the unauthorised transfer of a confidential document.

The examination was offline and used Volatility 2 on a read-only working copy in a Kali Linux virtual machine. In line with the case brief, no IP address, host, account, service or person named in the output was contacted, no recovered file was executed, no password or hash-cracking step was used, and the original memory image is not included in the evidence package.

Item	Detail
Evidence item	E01 to E20: memory image acquisition record, Volatility 2 output files and a supplementary strings search (Appendix D)
Scenario note	The scenario text describes a Windows 10 image. The supplied file is named memdumpWin7.mem and the output shows Windows 7 SP1, 32-bit with PAE (kernel 6.1.7601; console banner “Microsoft Windows [Version 6.1.7601]”). All findings use the profile the image supports.
Workstation indicators	User profile IEUser (SID ending -1000) with a second profile sshd_server; local IPv4 address 192.168.56.8; VirtualBox guest services (VBoxService, VBoxTray) and Hyper-V integration services (vmicsvc) running
Examiner	Mary-Joy Adewole, 2025/FWSD/11468

2. Executive Conclusion

The memory image was captured on 6 January 2019 at 15:09:06 UTC while a user session named IEUser was active. Volatility output from several independent plugins supports the following sequence:

1. **A browser session in the IEUser profile asked for the file.** Chrome (PID 2388) started at 15:04:30 UTC. Strings in memory show Chrome address-bar and search text containing http://192.168.56.5/secret_file.docx and a partial download path C:\Users\IEUser\Downloads\secret_file.docx.crdownload. Two closed TCP connections from 192.168.56.8 to 192.168.56.5 on port 80 are also recorded (owner process unknown).
2. **A USB flash drive was known to Windows as F:.** The USBSTOR, DeviceClasses, MountedDevices and MountPoints2 records all describe one “General UDisk USB Device”, assigned drive letter F: and registered under IEUser by 15:03:07 UTC.
3. **A command prompt copied the file to F:.** cmd.exe (PID 3996, started 15:06:11 UTC by explorer.exe) holds the history ipconfig, cd Downloads, copy secret_file.docx F:, and its screen buffer shows “1 file(s) copied.” at the C:\Users\IEUser\Downloads prompt.
4. **File Explorer recorded F:\ as the most recently used volume** (ShellBags, 15:07:03 UTC), 52 seconds after the command prompt started.

Read together, this supports a **probable completed copy of a file named secret_file.docx from the IEUser Downloads folder to a USB flash drive** a few minutes before the memory capture, where the file had been requested by Chrome from 192.168.56.5. It does not prove that the file was confidential, that its contents were unchanged, that the drive left the premises, or who was at the keyboard.

Confidence. Moderate to high that the copy command ran and reported success (command history and screen buffer agree). Moderate that the file came from 192.168.56.5 through Chrome (strings and netscan are consistent, but the strings are not tied to a process and the connections have no owner). Low for attributing the activity to a named person: the evidence identifies a profile (IEUser) and a workstation, not an individual. Section 7 lists the limitations and alternative explanations.

3. Evidence Acquisition and Integrity (Question 1)

The supplied memory image was copied from the course material (held in the VMware shared folder) into a case folder, hashed immediately, made read-only, and only then was a working copy taken. All analysis used the working copy. The record below is taken from the terminal output in Figures 1 to 3 and the saved files in `evidence_records/`.

Attribute	Value
Original file name	memdumpWin7.mem
Source	ICDFA SBT-DF204 Case Study 3 material, downloaded to Windows and read from the VMware shared folder `/mnt/hgfs/kali_documents/memdumpWin7.mem` (Figure 1)
Copied to	`~/SBTDF204_CS3/original/memdumpWin7.mem` using `cp -p` (preserves timestamps)
File size	1,073,676,288 bytes (`ls -l`, `stat`)

Attribute	Value
Acquisition date/time	Sunday 18 October 2026, 17:36:42 UTC (file birth time, converted with `date -u`; the Kali VM clock is UTC-4, so `stat` shows 13:36:42 -0400). Hashes re-verified at 17:46:30 UTC.
SHA-256	500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e
MD5	ac7eaa46a7006d5eb13cda13ee0b9bc9
File type (`file`)	data (a raw memory dump has no file-format header; the type is established by Volatility, not by `file`)
Original protection	`chmod a-w original/memdumpWin7.mem`; permissions afterwards `-r-xr-xr-x`
Working copy	`cp -p original/memdumpWin7.mem working/memdumpWin7.mem`; SHA-256 identical to the original ("HASHES MATCH", Figure 2). The working copy is also read-only: re-running the copy printed "Permission denied" and changed nothing.
Tool version	Volatility Foundation Volatility Framework 2.6.1, run as `python2 vol.py` from `~/volatility2` (Figure 4)
Precautions	Original never opened by an analysis tool; no upload of the image; offline examination; no recovered file executed; Volatility reads the image read-only. Output files were written to `evidence_records/` and listed in a SHA-256 manifest (E20).



Figure 1: Source file listed in the VMware shared folder (ls -l /mnt/hgfs/kali_documents/ | grep -i -E "mem|vol"): memdumpWin7.mem, 1,073,676,288 bytes. (E01)



Figure 2: Acquisition block: cp -p, date -u, stat, ls -l, sha256sum, md5sum, file, chmod a-w, working-copy cp -p and hash comparison. This is the re-run; the two "Permission denied" lines show the read-only copies were not overwritten. (E01)



Figure 3: acquisition_date_utc.txt: original copied 17:36:42 UTC (file birth time), hashes re-verified 17:46:30 UTC. (E01)

Note on the date record. The first acquisition block printed the current time, not the copy time, when it was re-run. The file birth time was therefore used to record when the original was actually copied (17:36:42 UTC), with the re-verification time kept separately (17:46:30 UTC).

4. Method and Profile Selection (Question 2)

Each stage was checked before the next began. Every Volatility command was run against the working copy, filtered with `grep -v "Failed to import"` (to hide plugin import messages for modules this case does not use) and saved with `tee` to a file named after the plugin and profile in `evidence_records/` (Appendix E). Volatility 2.6.1 was cloned from the Volatility Foundation repository; the shellbags, getsids and related plugins failed to import until `pycryptodome` was installed for Python 2 (Figure 4). The two plugins that still fail to import (apihooks and threads, which need `distorm3`) are not used in this case.

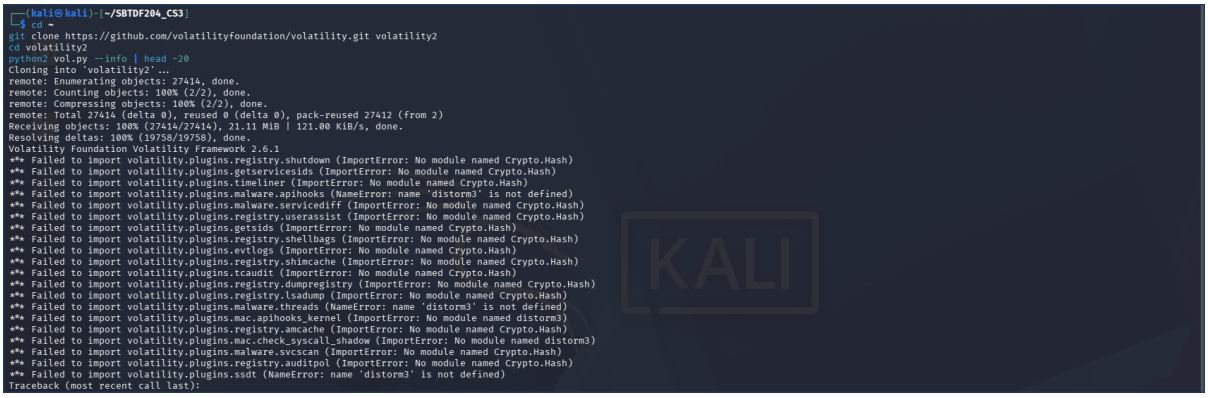


Figure 4: Volatility 2.6.1 cloned from the Volatility Foundation repository. The import failures (Crypto.Hash, distorm3) were resolved before analysis by installing pycryptodome for Python 2. (E02)

4.1 imageinfo result

`imageinfo` was run first, without a profile, so that the profile comes from the image and not from an assumption (Figure 5).

Attribute	Value
Suggested profiles	Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
Address space	IA32PagedMemoryPae (kernel) over FileAddressSpace; PAE type PAE; DTB 0x185000
KDBG / KPCR	KDBG 0x82972c68; KPCR for CPU 0: 0x82973d00; KUSER_SHARED_DATA 0xffdf0000; 1 processor
Image type	Service Pack 1
Image date and time (UTC)	**2019-01-06 15:09:06 UTC+0000**
Image local date and time	2019-01-06 07:09:06 -0800 (UTC-8, consistent with US Pacific standard time)



Figure 5: imageinfo, no profile: python2 vol.py -f working/memdumpWin7.mem imageinfo. Candidate profiles, KDBG, image date and time in UTC and local time. (E02)

4.2 Profile validation and selection

The brief requires a suggested profile to be validated with usable output from several plugins. The candidates were tested as follows.

Check	Result	Evidence
kdbgscan	The KDBG at physical offset 0x2972c68 passes the owner-tag check (True) and reports version 15.7601. All four candidate profiles are suggested for it, so the kernel is Windows 7 SP1 (build 7601). Two further KDBG hits (0xd6c7c68, 0xec94c68) are stale copies in RAM with the same pointers.	E03, Figure 6
pslist, Win7SP1x86_23418	57 lines; names, PIDs, parents and creation times form a consistent tree (System, smss, csrss, wininit, services, lsass, svchost, explorer, chrome).	E04, Figure 7, Figure 8
pslist, Win7SP1x86_24000	57 lines; `diff` against the 23418 run reports “pslist outputs IDENTICAL”. The choice between these two profiles therefore does not change any process result.	E04, Figure 9
Win7SP0x86	Not selected: imageinfo reports Image Type (Service Pack) 1.	E02
Win7SP1x86 (generic)	Same SP1 x86 structures; the build-specific profile was preferred.	E02, E03
psscan, netscan, cmdscan, consoles, hivelist, getsids, printkey, shellbags	All returned readable, mutually consistent output under Win7SP1x86_23418 (for example the IPv4 address 192.168.56.8 appears in both netscan and the console screen buffer, and explorer.exe PID 2524 is the parent of cmd.exe, chrome.exe and FTK Imager.exe in both pslist and psscan).	E05 to E17

Selected profile: `Win7SP1x86_23418`. It is the first-ranked suggestion for a verified KDBG, matches the Service Pack reported by imageinfo, and produced coherent output from every plugin used. I did not run netscan under the 24000 profile; the two profiles were compared on pslist only.

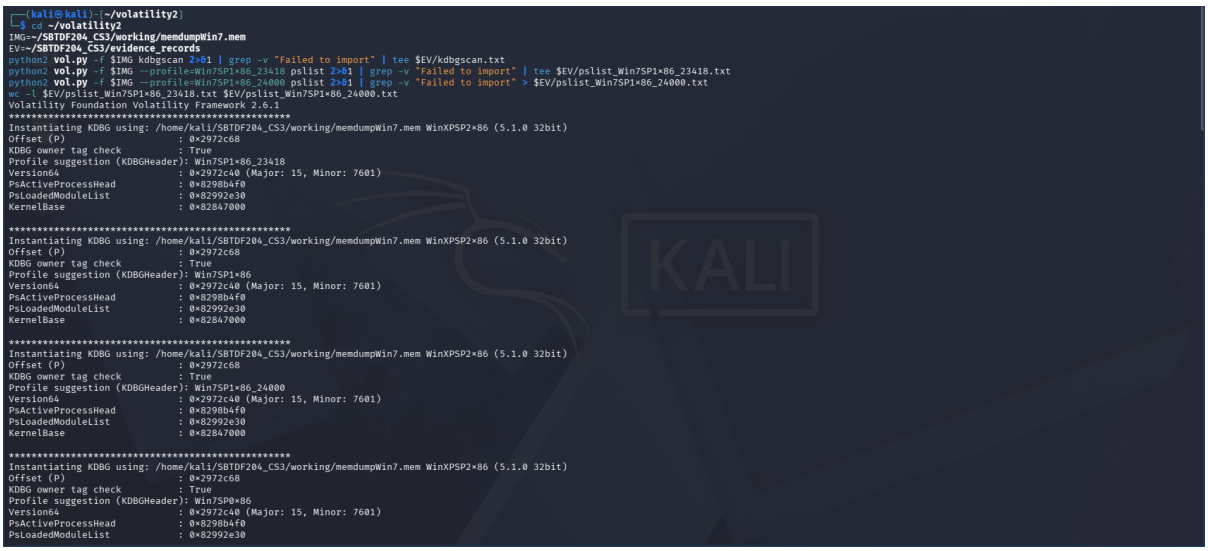


Figure 6: kdbgscan: KDBG owner-tag check True, version 15.7601; suggestions Win7SP1x86_23418, Win7SP1x86, Win7SP1x86_24000, Win7SP0x86. (E03)

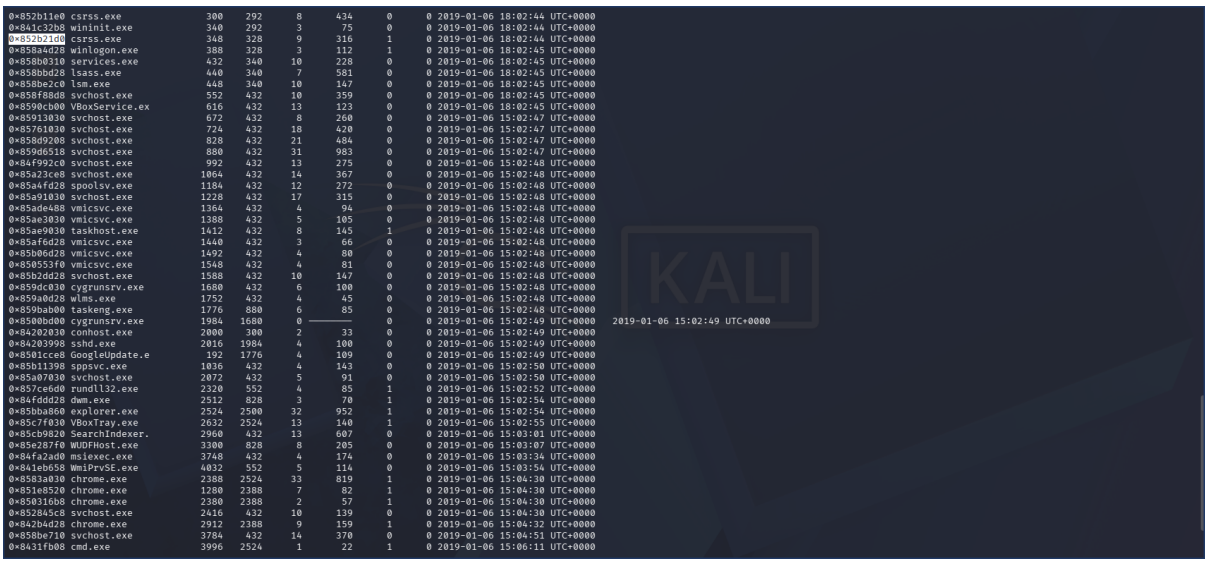


Figure 7: pslist, profile Win7SP1x86_23418 (saved to pslist_Win7SP1x86_23418.txt): coherent process tree. (E04)

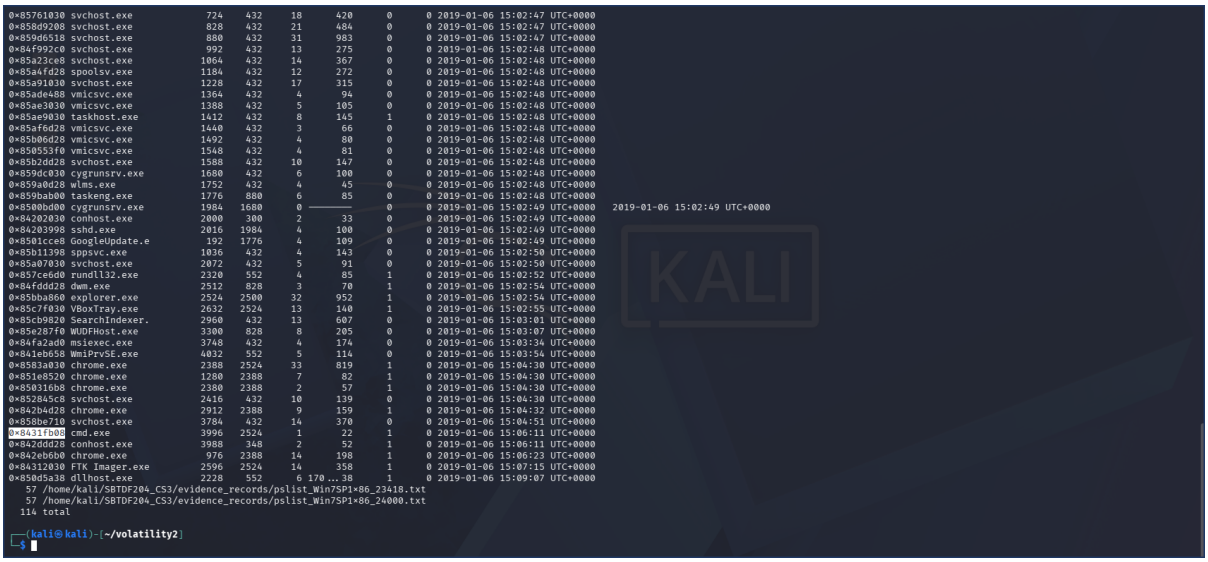


Figure 8: pslist, profile Win7SP1x86_23418, end of output (cmd.exe, conhost.exe, chrome.exe, FTK Imager.exe, dllhost.exe) and line counts for both profile runs (57 lines each). (E04)

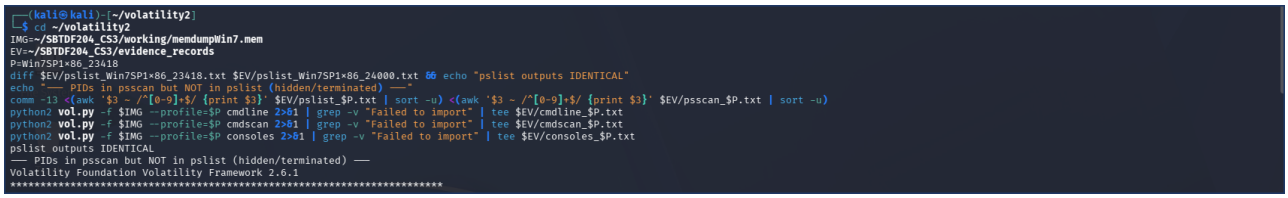


Figure 9: diff of the pslist output for Win7SP1x86_23418 and Win7SP1x86_24000 ("pslist outputs IDENTICAL") and the psscan-versus-pslist PID comparison (nothing listed). (E04, E05)

4.3 Image time, time zone and clock behaviour

- **Image time.** 2019-01-06 15:09:06 UTC is when the capture read the system time. Every artefact in the image is no later than this, apart from the small smear described in Section 7.
- **Time zone.** imageinfo gives a local time of UTC-8. Volatility prints process, network and registry key times with a UTC+0000 label, and this report uses them as printed. Local time is never added to the timeline.
- **Clock anomaly.** The ten earliest system processes (System, smss, both csrss, wininit, winlogon, services, lsass, lsm, svchost 552, VBoxService) show start times of 18:02:44 to

18:02:45 UTC, which is later than the image time. Every later process starts between 15:02:47 and 15:09:07. The offset is exactly 3 hours, so the guest clock was most probably corrected early in boot by a time-synchronisation service (VBoxService and the Hyper-V TimeSync service `vmicsvc -feature TimeSync` are both running). That is an inference. Those early start times are treated as unreliable, and the 15:02:47 onward times as consistent.

- **Scenario mismatch.** The scenario says Windows 10; the image is Windows 7 SP1 x86. The profile and every finding follow the image.

5. Findings

5.1 User and account artefacts (Question 3)

`hivelist` (E10) and `getsids` (E11) were used to identify the profiles on the machine and the account that owned the relevant processes (Figure 10 and Figure 11).

Item	Value
Interactive account	IEUser, SID S-1-5-21-1716914095-909560446-1177810406-1000
Hive paths (virtual offset)	NTUSER.DAT `\??\C:\Users\IEUser\ntuser.dat` (0x8b5d39b0); UsrClass.dat `\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat` (0xa620b008)
Second profile	sshd_server: NTUSER.DAT (0xa65269c8) and UsrClass.dat (0xa6539260). This is the OpenSSH service account; sshd.exe (PID 2016) is listening on port 22.
System hive	`\REGISTRY\MACHINE\SYSTEM` (0x87c1c008), used for USBSTOR, MountedDevices and DeviceClasses
Process ownership	explorer.exe (2524), cmd.exe (3996) and FTK Imager.exe (2596) all run as IEUser. explorer.exe and cmd.exe are at Medium mandatory level (S-1-16-8192); FTK Imager.exe is at High (S-1-16-12288), i.e. elevated.
Group membership shown	Administrators (S-1-5-32-544), Users, Interactive (S-1-5-4), Console Logon (S-1-2-1), Local account (S-1-5-113) and Logon Session S-1-5-5-0-61708 for the IEUser processes

What this shows. The commands and Explorer activity examined below belong to the IEUser profile, and that session is an interactive one (Console Logon SID). **What it does not show.** A profile is not a person. IEUser is a generic account name, so the name alone does not point to one individual. The machine also has a remote-access service (OpenSSH, with its own sshd_server profile), which would be an alternative route to the same profile. No logon event or authentication record is present in this evidence set to say who was at the keyboard.



Figure 10: hivelist (top) and getsids for explorer.exe (2524) and cmd.exe (3996), profile Win7SP1x86_23418. (E10, E11)



Figure 11: getsids for cmd.exe (3996) and FTK Imager.exe (2596): same user SID; FTK Imager.exe runs at High mandatory level. (E11)

5.2 Process and network evidence (Question 4)

Processes

`pslist` and `psscan` (E04, E05) were compared by PID: the check printed nothing, so no process in `psscan` is missing from `pslist` (Figure 9). The only terminated process is cygrunsrv.exe PID 1984, which exited at 15:02:49 UTC, the same second it started, and spawned sshd.exe (PID 2016). The processes that matter to the case are:

PID	Process	Parent	Created (UTC)	Note
2524	explorer.exe	2500	15:02:54	IEUser desktop shell; parent of the three processes below
2388	chrome.exe	2524	15:04:30	Browser main process. Children: 2380 and 1280 (15:04:30), 2912 (15:04:32), renderer 976 (15:06:23), command lines in Figure 14
3996 / 3988	cmd.exe / conhost.exe	2524 / 348	15:06:11	Interactive command prompt and its console host

PID	Process	Parent	Created (UTC)	Note
2596	FTK Imager.exe	2524	15:07:15	Memory-capture tool (run from C:\Program Files\AccessData\FTK Imager\); High integrity
2016	sshd.exe	1984	15:02:49	OpenSSH daemon, listening on port 22; no connection to it is recorded
2228	dllhost.exe	552	15:09:07	Created one second after the image time (capture smear)

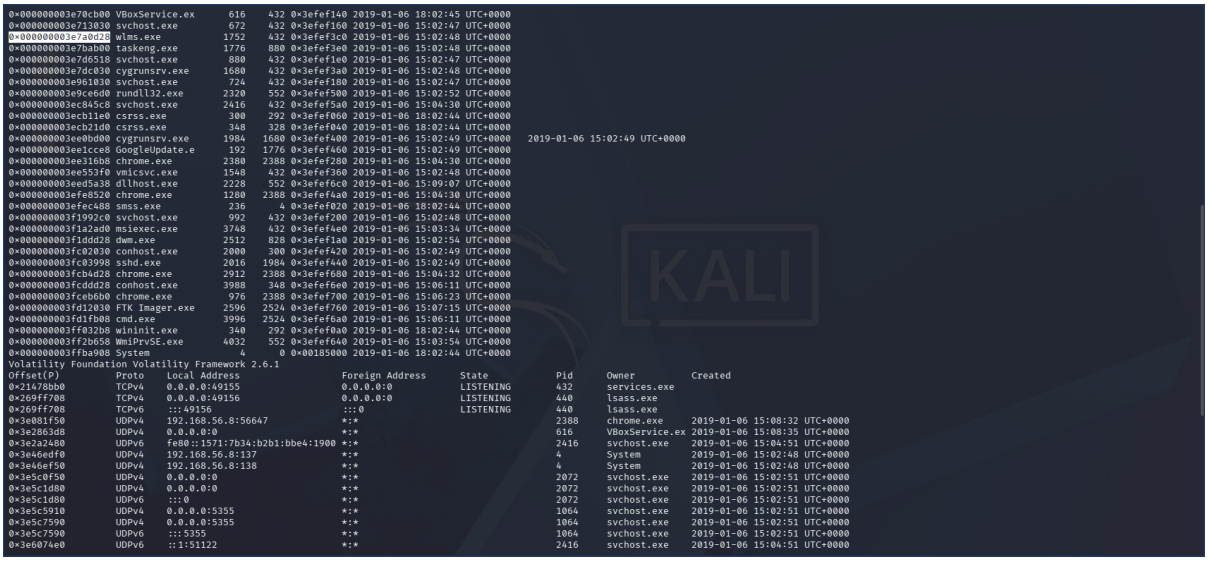


Figure 12: psscan (end of output) and the header of netscan, profile Win7SP1x86_23418. (E05, E06)

Network

`netscan` (E06) lists no ESTABLISHED connection at capture time. The local address is 192.168.56.8 (a private host-only style network). The entries that bear on the case are:

Offset (P)	Proto / local	Foreign	State	PID / owner	Created
0x3e043b10	TCPv4 192.168.56.8:49177	192.168.56.5:80	CLOSED	-1 (none)	not recorded
0x3fcba008	TCPv4 192.168.56.8:49178	192.168.56.5:80	CLOSED	-1 (none)	not recorded
0x3e081f50	UDPv4 192.168.56.8:56647	*:*	(UDP)	2388 chrome.exe	15:08:32 UTC
0x3ee1aa88	TCPv4 0.0.0.0:22	0.0.0.0:0	LISTENING	2016 sshd.exe	not recorded

The two closed connections are consecutive ephemeral ports to port 80 (plain HTTP) on 192.168.56.5, the same address that appears in the Chrome strings below. Their owner is PID -1, meaning the owning process structure was gone or unreadable, so they **cannot be attributed to Chrome or any process**, and no creation time is recorded. A closed entry shows that a connection structure existed, not that data was transferred. No endpoint was contacted.



```
Offset(P)          Proto    Local Address                  Foreign Address        State          Pid      Owner          Created
0x21478bb8         TCPv4    0.0.0.0:49155                  0.0.0.0:0              LISTENING      432      services.exe
0x269ff708         TCPv4    0.0.0.0:49156                  0.0.0.0:0              LISTENING      440      lsass.exe
0x269ff708         TCPv6    :::49156                       :::0                   LISTENING      440      lsass.exe
0x3e081f50         UDPv4    192.168.56.8:56647             *:*                                   2388     chrome.exe     2019-01-06 15:08:32 UTC+0000
0x3e2863d8         UDPv4    0.0.0.0:0                      *:*                                   616      VBoxService.ex 2019-01-06 15:08:35 UTC+0000
0x3e2a2480         UDPv6    fe80::1571:7b34:b2b1:bbe4:1900 *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e46ed0         UDPv4    192.168.56.8:137               *:*                                   4        System         2019-01-06 15:02:48 UTC+0000
0x3e46ef50         UDPv4    192.168.56.8:138               *:*                                   4        System         2019-01-06 15:02:48 UTC+0000
0x3e5c0f50         UDPv4    0.0.0.0:0                      *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv4    0.0.0.0:0                      *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv6    :::0                           *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c5910         UDPv4    0.0.0.0:5355                   *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv4    0.0.0.0:5355                   *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv6    :::5355                        *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e6074e0         UDPv6    ::1:51122                      *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e7f2828         UDPv4    0.0.0.0:0                      *:*                                   1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e7f2828         UDPv6    :::0                           *:*                                   1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e446008         TCPv4    0.0.0.0:49154                  0.0.0.0:0              LISTENING      880      svchost.exe
0x3e446d98         TCPv4    0.0.0.0:49154                  0.0.0.0:0              LISTENING      880      svchost.exe
0x3e446d98         TCPv6    :::49154                       :::0                   LISTENING      880      svchost.exe
0x3e4e0c0          TCPv4    192.168.56.8:139               0.0.0.0:0              LISTENING      4        System
0x3e671988         TCPv4    0.0.0.0:445                    0.0.0.0:0              LISTENING      4        System
0x3e671988         TCPv6    :::445                         :::0                   LISTENING      4        System
0x3e72e530         TCPv4    0.0.0.0:135                    0.0.0.0:0              LISTENING      672      svchost.exe
0x3e72ee48         TCPv4    0.0.0.0:135                    0.0.0.0:0              LISTENING      672      svchost.exe
0x3e72ee48         TCPv6    :::135                         :::0                   LISTENING      672      svchost.exe
0x3e736388         TCPv4    0.0.0.0:49152                  0.0.0.0:0              LISTENING      340      wininit.exe
0x3e736d98         TCPv4    0.0.0.0:49152                  0.0.0.0:0              LISTENING      340      wininit.exe
0x3e736d98         TCPv6    :::49152                       :::0                   LISTENING      340      wininit.exe
0x3e79d528         TCPv4    0.0.0.0:49153                  0.0.0.0:0              LISTENING      724      svchost.exe
0x3e79d528         TCPv6    :::49153                       :::0                   LISTENING      724      svchost.exe
0x3e7a1618         TCPv4    0.0.0.0:49153                  0.0.0.0:0              LISTENING      724      svchost.exe
0x3e043b10         TCPv4    192.168.56.8:49177             192.168.56.5:80        CLOSED         -1
0x3ec6c820         UDPv4    127.0.0.1:51123                *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3f19a820         UDPv6    ::1:1900                       *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e8f0620         TCPv4    0.0.0.0:49156                  0.0.0.0:0              LISTENING      440      lsass.exe
0x3e9b3e50         TCPv4    0.0.0.0:49155                  0.0.0.0:0              LISTENING      432      services.exe
0x3e9b3e50         TCPv6    :::49155                       :::0                   LISTENING      432      services.exe
0x3ee1aa88         TCPv4    0.0.0.0:22                     0.0.0.0:0              LISTENING      2016     sshd.exe
0x3ee1bc18         TCPv4    0.0.0.0:22                     0.0.0.0:0              LISTENING      2016     sshd.exe
0x3ee1bc18         TCPv6    :::22                          :::0                   LISTENING      2016     sshd.exe
0x3fc0e008         UDPv4    127.0.0.1:1900                 *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fc95d48         UDPv4    192.168.56.8:1900              *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fcba008         TCPv4    192.168.56.8:49178             192.168.56.5:80        CLOSED         -1

┌──(kali㉿kali)-[~/volatility2]
└─$
```

Figure 13: netscan, profile Win7SP1x86_23418: listeners, the Chrome UDP socket and the two CLOSED TCP connections to 192.168.56.5:80 with owner PID -1. (E06)

Command lines and file objects

`cmdline` (E07) shows cmd.exe as plain `C:\Windows\system32\cmd.exe` and FTK Imager.exe from its install path (Figure 14). `filescan` (E18) found no file object for `secret_file.docx`: none of the 2,006 lines of output contains “secret” or “.docx”. It did list `\Device\HarddiskVolume1\Users\IEUser\Desktop\memdumpWin7.mem` (the capture file being written), …`\IEUser\Downloads` and …`\IEUser\Links\Downloads.lnk`, Chrome Download Service database files, and two directory objects for the root of `\Device\HarddiskVolume2\`, a second volume besides the system volume (Figure 15). The output does not show which drive letter HarddiskVolume2 has, so that entry is noted but not relied on. The Chrome files show that the Download Service was active; they are not a download record.



```
142:chrome.exe pid:   2912
143-Command line :
--
148:cmd.exe pid:   3996
149-Command line : "C:\Windows\system32\cmd.exe"
--
151:conhost.exe pid:   3988
152-Command line : \??\C:\Windows\system32\conhost.exe "-164647720-2759156771236404584117504074 2-950872444-977965776-107363114-188091992
--
154:chrome.exe pid:    976
155-Command line : "C:\Program Files\Google\Chrome\Application\chrome.exe" --type=renderer --field-trial-handle=904,17926919778621107 08,4827585219111034411,131072 --disable-gpu-compositing --service-pipe-token=
4706400379687013043 --lang=en-US --enable-offline-auto-reload --enable-offline-auto-reload-visible-only --device-scale-factor=1 --num-raster-threads=1 --service-request-channel-token=4706400379687013043 --rende
rer-client-id=13 --mojo-platform-channel-handle=3460 /prefetch:1
--
157:FTK Imager.exe pid:   2596
158-Command line : "C:\Program Files\AccessData\FTK Imager\FTK Imager.exe"
3:CommandProcess: conhost.exe Pid: 2000
4:CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
8:Cmd #29 @ 0x3200c4: 374227272
9:Cmd #37 @ 0x3200c4: 374227272
11:CommandProcess: conhost.exe Pid: 3988
12:CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
16:Cmd #0 @ 0x10dc40: ipconfig
17:Cmd #1 @ 0x107f90: cd Downloads
18:Cmd #2 @ 0x1147a0: copy secret_file.docx F:
19:Cmd #22 @ 0xff818488: ?
20:Cmd #25 @ 0xff818488: ?
21:Cmd #36 @ 0xe00c4: ?????
22:Cmd #37 @ 0x10cff0: ?????
Volatility Foundation Volatility Framework 2.6.1
Virtual    Physical   Name
---------- ---------- ----
0x87c10370 0x27fd5370 [no name]
0x87c1c008 0x27fa3008 \REGISTRY\MACHINE\SYSTEM
0x87c459c8 0x27d8e9c8 \REGISTRY\MACHINE\HARDWARE
0x889c0430 0x1285a430 \??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0x8b46d008 0x16d8c008 \??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0x8b5d39b0 0x120049b0 \??\C:\Users\IEUser\ntuser.dat
0x97104008 0x1c45f008 \SystemRoot\System32\Config\SECURITY
0x9711d9c8 0x1e6a09c8 \SystemRoot\System32\Config\SOFTWARE
0x981f9008 0x1b446008 \SystemRoot\System32\Config\DEFAULT
0x98209518 0x1f3a7518 \SystemRoot\System32\Config\SAM
0x982d55c0 0x1e17c5c0 \REGISTRY\MACHINE\BCD00000000
0xa620b008 0x1997b008 \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa63269c8 0x17ad09c8 \??\C:\Users\sshd_server\ntuser.dat
0xa6539260 0x0f690260 \??\C:\Users\sshd_server\AppData\Local\Microsoft\Windows\UsrClass.dat
0xa7d2a9c8 0x01c469c8 \??\C:\System Volume Information\Syscache.hve
```

Figure 14: cmdline for cmd.exe (3996), conhost.exe (3988), chrome.exe (976) and FTK Imager.exe (2596); cmdscan lines; hivelist. Profile Win7SP1x86_23418. (E07, E08, E10)

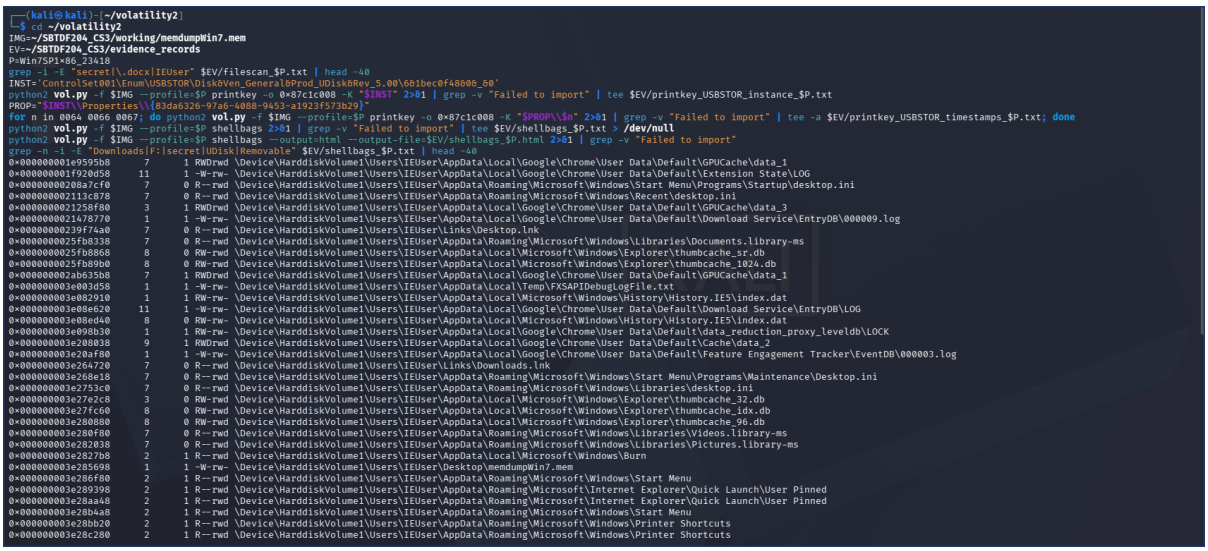


Figure 15: filescan filtered for secret|.docx|IEUser, profile Win7SP1x86_23418: Desktop\memdumpWin7.mem, Links\Downloads.lnk and Chrome Download Service files; no secret_file.docx object. (E18)

Memory strings (supplementary)

A strings search of the working copy for `secret_file` (E19, not a Volatility plugin and not tied to a process) found, in ASCII and UTF-16 text (Figure 16):

- Chrome address-bar and search text containing `http://192.168.56.5/secret_file.docx` and the Google-search form of the same text (`google.com/search?q=192.168.56.5%2Fsecret_file.docx`), plus many shorter prefixes of it (…`secret_file.d`, …`secret_file.do`) and a few mistyped addresses (for example 192.128.56.5 and 196.128.56.5).
- `C:\Users\IEUser\Downloads\secret_file.docx.crdownload` and a `.crdownload…tmp` name (Chrome's temporary name while a download is in progress), `file:C:/Users/IEUser/Downloads/secret_file.docx`, `C:\Users\IEUser\Downloads\secret_file.docx` and `F:\secret_file.docx`.
- The same cmd.exe screen text as in Section 5.3.

These strings are consistent with Chrome being asked for the file over HTTP from 192.168.56.5 and writing it into the IEUser Downloads folder, followed by a copy to F:. The incremental prefixes suggest the address was entered by keyboard rather than clicked. That is an inference. A string in RAM has no timestamp and no process owner, a `.crdownload` name shows that a download started and not that it finished, and `F:\secret_file.docx` shows a path, not a completed write. The later copy command uses the final name `secret_file.docx`, so a file under that name existed in Downloads when the command ran.

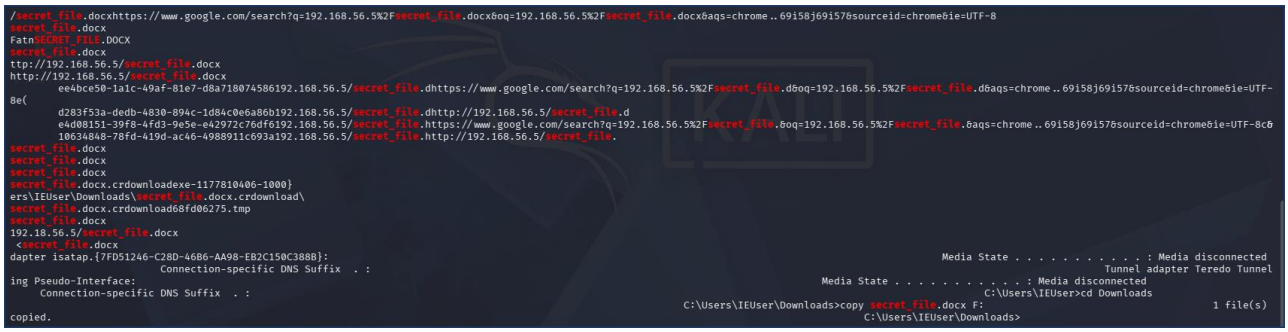


Figure 16: strings (ASCII and UTF-16) on the working copy filtered for secret_file. Supplementary, not a Volatility plugin: Chrome address-bar text, the URL, the .crdownload path and console text. (E19)

5.3 Command history (Question 5)

`cmdscan` (E08) and `consoles` (E09) both recover the history of the interactive command prompt. They agree (Figure 14 and Figure 17).

Console / process	History	Screen buffer
conhost.exe 3988 with cmd.exe 3996 (IEUser)	Cmd #0 `ipconfig`; Cmd #1 `cd Downloads`; Cmd #2 `copy secret_file.docx F:` (CommandCount 3, LastAdded 2)	Prompt `C:\Users\IEUser>` then `ipconfig` output (IPv4 192.168.56.8, mask 255.255.255.0, no default gateway), `cd Downloads`, then `C:\Users\IEUser\Downloads>copy secret_file.docx F:` followed by `1 file(s) copied.`
conhost.exe 2000 with sshd.exe 2016 (cygrunsrv console)	CommandCount 0 (empty)	Empty

Reading. The prompt shows that the commands ran in the IEUser profile from the Downloads folder. The ipconfig output ties the console to the same host address seen in netscan. The line `1 file(s) copied.` directly under the copy command is the command's own success message, which is stronger than a command string alone.

Limits.

- The history and the screen buffer carry no timestamps. The commands can only be placed between the start of cmd.exe (15:06:11 UTC) and the image time (15:09:06 UTC).
- The success message does not show the file's size, content or hash, or what the file actually was. It does not show that the destination `F:` was the USB device at that moment; that comes from the registry records in Section 5.4.
- `cmdscan` also printed corrupted fragments for the sshd console (`Cmd #29` and `Cmd #37` with unreadable text) and for the cmd.exe buffer (`Cmd #22`, `#25`, `#36`, `#37` with `?` or unreadable bytes). These are damaged or reused memory, not recoverable commands, and are not interpreted.
- Console history is held per console session. Commands typed in another console, or before this one started, would not appear.



Figure 17: consoles, profile Win7SP1x86_23418: cmd.exe (3996) history and screen buffer: ipconfig, cd Downloads, copy secret_file.docx F: and "1 file(s) copied." (E09)

5.4 Removable-media evidence (Question 6)

Registry keys were read from the in-memory hives with `printkey -o <hive offset> -K <key>` (E12 to E16). The SYSTEM hive is at 0x87c1c008 and the IEUser NTUSER.DAT at 0x8b5d39b0 (Section 5.1). The `Select` key shows `Current = 1`, so `ControlSet001` is the live control set (Figure 21).

Artefact	Record	Last write (UTC)	What it shows	What it does not show
USBSTOR (E14)	`Disk&Ven_General&Prod_UDisk&Rev_5.00`, instance `6&1bec0f48&0&_&0`; FriendlyName “General UDisk USB Device”; ContainerID {421e7e37-11dd-11e9-b3cf-806e6f6e6963}	15:02:54 (device and instance keys)	A generic USB flash disk was enumerated by Windows	A unique drive: the instance ID begins `6&` with `&` as its second character, the form Windows generates when a device has no serial number. First-connect and last-removal times (Properties 0064, 0066, 0067) were not recoverable.
DeviceClasses (E15)	`##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}`	15:02:54	The disk interface for this device exists	Who connected it, or when it was last plugged in
MountedDevices (E12)	`\DosDevices\F:` decodes to `\??\USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56307-…}`; volume {421e7e52-11dd-11e9-b3cf-08002709e15d}. `C:` is a disk signature; `E:` is the VirtualBox CD-ROM.	15:02:54	Drive letter F: is mapped to the USB disk	That the mapping applied at the moment of the copy (the key time predates the command)
MountPoints2, IEUser NTUSER.DAT (E13)	Subkeys `CPC`, `F`, `{421e7e52-11dd-11e9-b3cf-08002709e15d}` and four older volume GUIDs	15:03:07	The IEUser shell registered the F: volume by 15:03:07	Possession, or that the person who plugged it in is the account holder; the key time is the latest change, not the first or last use

Relationship. All four records point to the same USB disk, the F: drive letter and the IEUser profile. The `F:` target of `copy secret_file.docx F:` is therefore most likely this device. The two GUIDs (ContainerID and volume) decode, as time-based UUIDs, to 2019-01-06 18:02:40 UTC. That is 3 hours ahead of the image time and within seconds of the early-boot processes that carry the same offset (Section 4.3), so it reflects the uncorrected guest clock and is **not** used as a UTC event time.



Figure 18: printkey MountedDevices (SYSTEM hive 0x87c1c008): key last updated 2019-01-06 15:02:54 UTC. (E12)

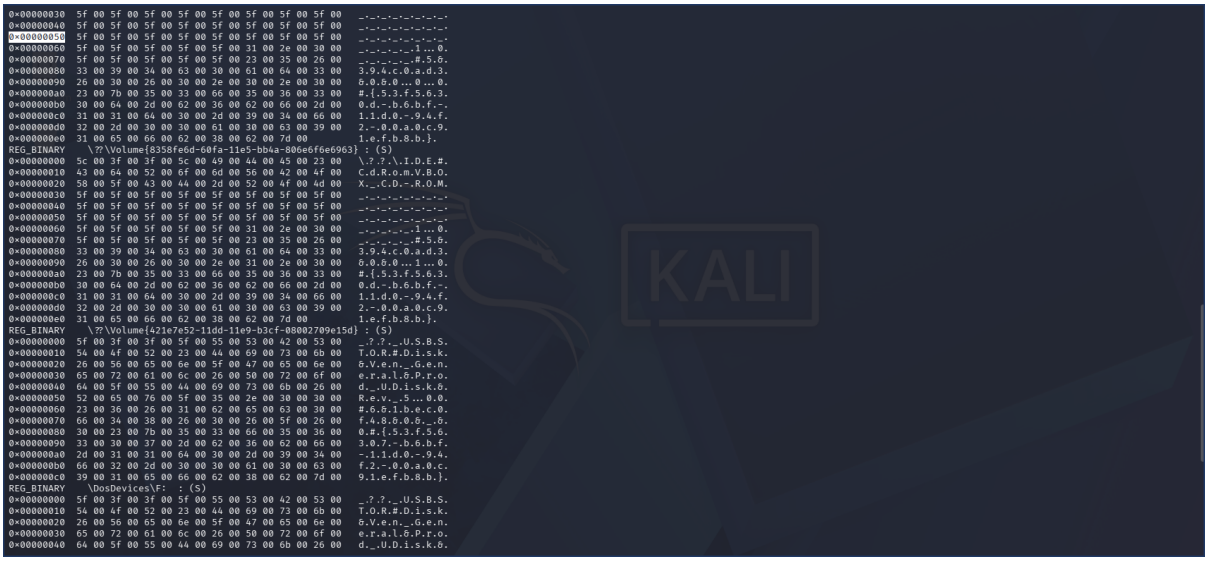


Figure 19: printkey MountedDevices: \DosDevices\F: decodes to USBSTOR#Disk&Ven_General&Prod_UDisk. (E12)

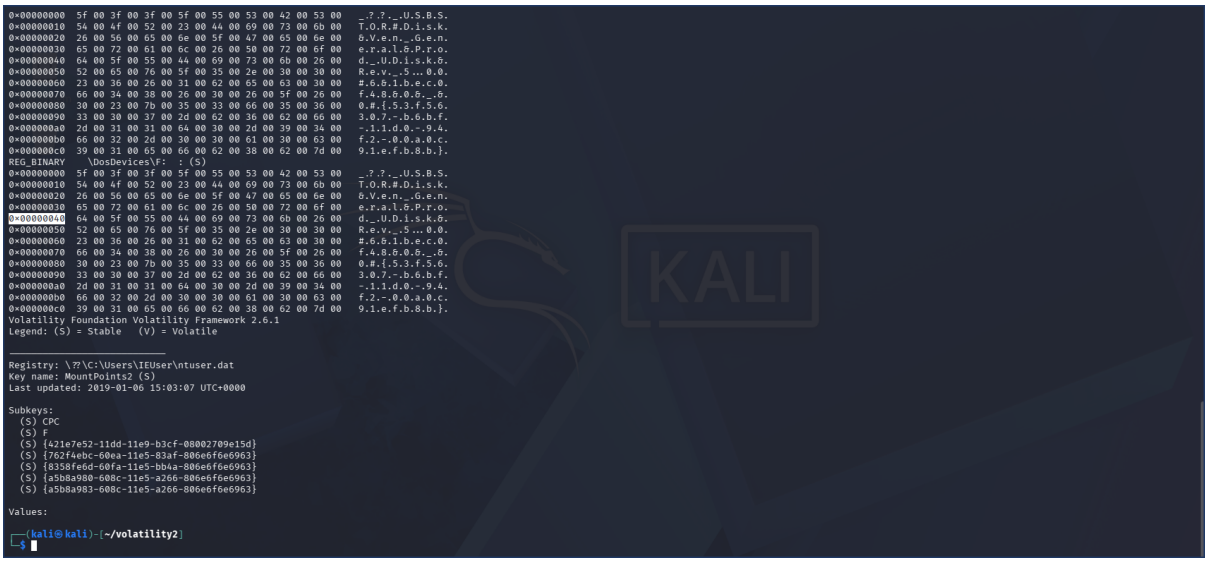


Figure 20: printkey MountPoints2 in the IEUser NTUSER.DAT (0x8b5d39b0): last updated 15:03:07 UTC; subkeys F and {421e7e52-…}. (E13)

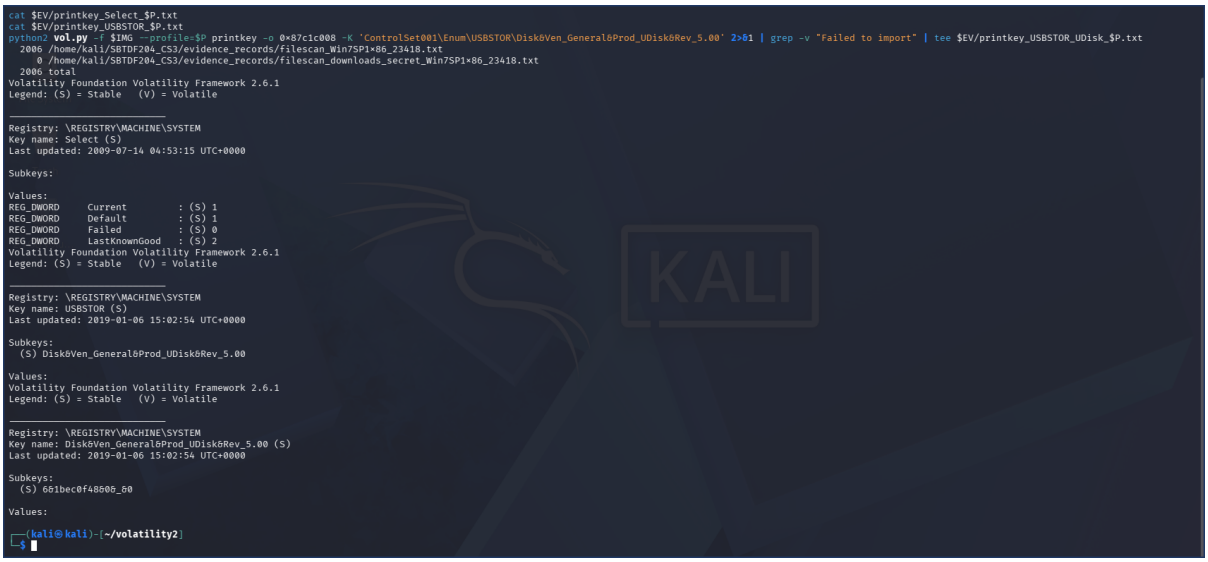


Figure 21: printkey Select (Current = 1, so ControlSet001 is live) and USBSTOR with its device-class key. (E14, E16)



Figure 22: printkey on the USBSTOR instance key 6&1bec0f48&0&_&0: FriendlyName, ContainerID, HardwareID. (E14)

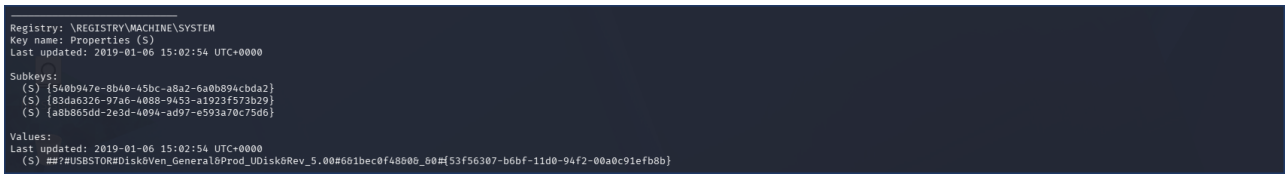


Figure 23: printkey on the instance Properties key and on DeviceClasses {53f56307-…}: last updated 15:02:54 UTC. (E14, E15)

5.5 File Explorer evidence: ShellBags (Question 7)

`shellbags` (E17) was run with the selected profile and saved as text and HTML. The entries that bear on the case are in the IEUser UsrClass.dat (Figure 24 to Figure 26).

Key (UsrClass.dat unless stated)	Last write (UTC)	Entries	Reading
`Local Settings\…\Shell\BagMRU\0`	15:07:03	Volume names C:\, A:\, D:\, E:\, F:\ with F:\ at MRU position 0	F:\ is the most recently used volume node. Folder-view evidence that Explorer under IEUser reached F:\, 52 seconds after cmd.exe started and 2 min 3 s before the image time.
`…\BagMRU\0\1\1\0`	15:04:40	Downloads (`DOWNLO~1`, path `C:\Users\IEUser\Downloads`) and AppData	Explorer folder views of Downloads and AppData were updated at 15:04:40, ten seconds after Chrome started.
`…\BagMRU\4`	14:55:45	Downloads known folder {374de290-123f-4565-9164-39c4925e467b}	Downloads was opened in Explorer at some point at or before 14:55:45, which is earlier than this boot session (see below).
NTUSER.DAT `Shell\Bags\1\Desktop`	14:32:53	Desktop shortcuts including Google Chrome.lnk and OSForensics.lnk	Desktop layout only; also earlier than this boot session.

Folder times. For the Downloads item in `BagMRU\0\1\1\0` Volatility shows Modified 2019-01-06 14:49:40, Created 2015-09-21 09:17:34 and Accessed 2019-01-06 14:49:40 (Figure 25). These are the folder's own stamps, stored in the shell item in DOS format. They are **not** the modification time of `secret_file.docx` and do not show when it was copied or opened. Two cautions apply: DOS stamps carry no time zone and are written in the machine's local time, whereas Volatility labels them UTC, and these values are earlier than every process of this boot session (the first reliable start is 15:02:47). They are consistent with the Downloads folder having

changed in an earlier session or under a different clock setting, so they are used only as ordering hints.

What ShellBags establish here. Explorer under the IEUser profile browsed the Downloads folder and the F:\ volume, with F:\ being the latest. **What they do not establish.** They record folder views and window preferences. They do not list file names that were copied, and a folder view is not a file transfer.



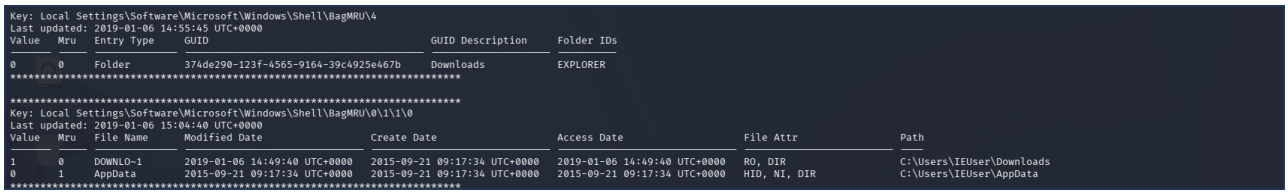
```
**************************************************************************
Registry: \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0
Last updated: 2019-01-06 15:07:03 UTC+0000
Value   Mru   Entry Type      Path
------  ----  --------------  ----
1       1     Volume Name     C:\
0       4     Volume Name     A:\
3       3     Volume Name     D:\
2       2     Volume Name     E:\
4       0     Volume Name     F:\
**************************************************************************

**************************************************************************
Registry: \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\2
Last updated: 2015-09-23 09:45:21 UTC+0000
Value   Mru   Entry Type      GUID                                    GUID Description     Folder IDs
------  ----  --------------  --------------------------------------  -------------------  ----------
0       0     Folder          7b0db17d-9cd2-4a93-9733-46cc89022e7c    Documents Library    EXPLORER
**************************************************************************

**************************************************************************
Registry: \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\4
Last updated: 2019-01-06 14:55:45 UTC+0000
Value   Mru   Entry Type      GUID                                    GUID Description     Folder IDs
------  ----  --------------  --------------------------------------  -------------------  ----------
0       0     Folder          374de290-123f-4565-9164-39c4925e467b    Downloads            EXPLORER
**************************************************************************

**************************************************************************
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1\0
Last updated: 2019-01-06 15:04:40 UTC+0000
```

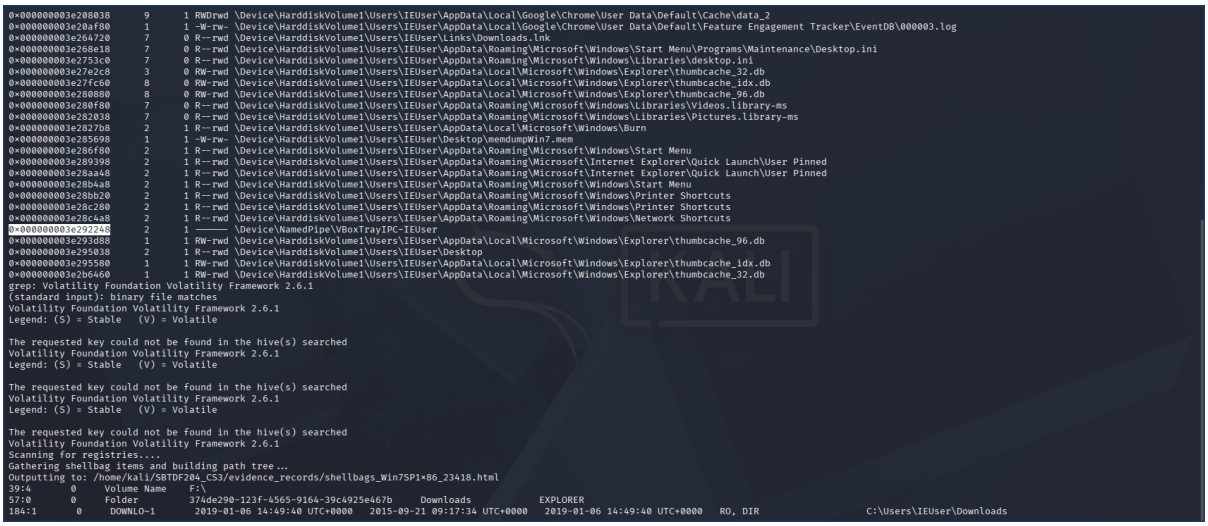
Figure 24: shellbags: BagMRU\0 in the IEUser UsrClass.dat lists volumes C:, A:, D:, E:, F:, with F:\ at MRU position 0 (key last updated 15:07:03 UTC). (E17)



```
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\4
Last updated: 2019-01-06 14:55:45 UTC+0000
Value   Mru   Entry Type      GUID                                    GUID Description     Folder IDs
------  ----  --------------  --------------------------------------  -------------------  ----------
0       0     Folder          374de290-123f-4565-9164-39c4925e467b    Downloads            EXPLORER
**************************************************************************

**************************************************************************
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1\0
Last updated: 2019-01-06 15:04:40 UTC+0000
Value   Mru   File Name       Modified Date                Create Date                  Access Date                  File Attr              Path
------  ----  --------------  ---------------------------  ---------------------------  ---------------------------  ---------------------  ----
1       0     DOWNLO~1        2019-01-06 14:49:40 UTC+0000  2015-09-21 09:17:34 UTC+0000  2019-01-06 14:49:40 UTC+0000  RO, DIR                C:\Users\IEUser\Downloads
0       1     AppData         2015-09-21 09:17:34 UTC+0000  2015-09-21 09:17:34 UTC+0000  2015-09-21 09:17:34 UTC+0000  HID, NI, DIR           C:\Users\IEUser\AppData
**************************************************************************
```

Figure 25: shellbags: Downloads bags (BagMRU\4 last updated 14:55:45 UTC; BagMRU\0\1\1\0 last updated 15:04:40 UTC) and the folder's own modified, created and accessed stamps. (E17)



```
0x000000003e208038      9      1 RWDrwd \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Google\Chrome\User Data\Default\Cache\data_2
0x000000003e20af80      1      1 -W-rw-  \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Google\Chrome\User Data\Default\Feature Engagement Tracker\EventDB\000003.log
0x000000003e264720      7      0 R--rwd \Device\HarddiskVolume1\Users\IEUser\Links\Downloads.lnk
0x000000003e26be18      7      0 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Maintenance\Desktop.ini
0x000000003e2753c0      7      0 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Libraries\desktop.ini
0x000000003e27e2c8      3      0 RW-rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Microsoft\Windows\Explorer\thumbcache_32.db
0x000000003e27fc60      8      0 RW-rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Microsoft\Windows\Explorer\thumbcache_idx.db
0x000000003e280880      8      0 RW-rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Microsoft\Windows\Explorer\thumbcache_96.db
0x000000003e280f80      7      0 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Libraries\Videos.library-ms
0x000000003e282038      7      0 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Libraries\Pictures.library-ms
0x000000003e2827b8      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Microsoft\Windows\Burn
0x000000003e285698      1      1 -W-rw-  \Device\HarddiskVolume1\Users\IEUser\Desktop\memdumpWin7.mem
0x000000003e286f80      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Start Menu
0x000000003e289398      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Internet Explorer\Quick Launch\User Pinned
0x000000003e28aa48      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Internet Explorer\Quick Launch\User Pinned
0x000000003e28b4a8      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Start Menu
0x000000003e28bb20      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Printer Shortcuts
0x000000003e28c280      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Printer Shortcuts
0x000000003e28c4a8      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Roaming\Microsoft\Windows\Network Shortcuts
0x000000003e29224c      2      1 ------  \Device\NamedPipe\VBoxTrayIPC-IEUser
0x000000003e293d88      1      1 RW-rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Microsoft\Windows\Explorer\thumbcache_96.db
0x000000003e295038      2      1 R--rwd \Device\HarddiskVolume1\Users\IEUser\Desktop
0x000000003e295580      1      1 RW-rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Microsoft\Windows\Explorer\thumbcache_idx.db
0x000000003e2b6460      1      1 RW-rwd \Device\HarddiskVolume1\Users\IEUser\AppData\Local\Microsoft\Windows\Explorer\thumbcache_32.db
grep: Volatility Foundation Volatility Framework 2.6.1
(standard input): binary file matches
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

The requested key could not be found in the hive(s) searched
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

The requested key could not be found in the hive(s) searched
Volatility Foundation Volatility Framework 2.6.1
Legend: (S) = Stable   (V) = Volatile

The requested key could not be found in the hive(s) searched
Volatility Foundation Volatility Framework 2.6.1
Scanning for registries....
Gathering shellbag items and building path tree...
Outputting to: /home/kali/SBTDF204_CS3/evidence_records/shellbags_Win7SP1x86_23418.html
39:4       0     Volume Name     F:\
57:0       0     Folder          374de290-123f-4565-9164-39c4925e467b     Downloads            EXPLORER
184:1      0     DOWNLO~1        2019-01-06 14:49:40 UTC+0000   2015-09-21 09:17:34 UTC+0000   2019-01-06 14:49:40 UTC+0000   RO, DIR                C:\Users\IEUser\Downloads
```

Figure 26: Saved shellbags text searched for F:, Downloads and related terms (lines 39, 57, 184); the three "requested key could not be found" results above it are the Properties time-stamp keys (0064, 0066, 0067), which were not recoverable. (E17, E14)

6. Timeline in UTC

Times are in UTC as printed by Volatility. Rows with no time are placed only where the evidence allows; undated rows are shown for completeness and are not part of the dated sequence. The full table with the source artefact for each row is in Appendix C.

Stage	UTC	Evidence	Observed fact and limit
Browser or process evidence	2019-01-06 15:04:30	E04, E05	pslist / psscan: chrome.exe 2388 (parent explorer.exe 2524); children 2380, 1280 (15:04:30), 2912 (15:04:32), renderer 976 (15:06:23) **Limit:** Chrome process tree created from the IEUser shell 4 min 36 s before image time. Process creation only: no URL, page or download is shown.
Browser or process evidence	Undated (RAM strings)	E19	strings: Chrome address-bar text `http://192.168.56.5/secret_file.docx` and prefixes of it; `C:\Users\IEUser\Downloads\secret_file.docx.crdownload`; `F:\secret_file.docx` **Limit:** Consistent with Chrome requesting the file and starting a download. No timestamp, no process owner (not Volatility output). `.crdownload` shows a download began, not that it finished.
Browser or process evidence	Undated (closed)	E06	netscan 0x3e043b10 and 0x3fcba008: TCPv4 192.168.56.8:49177 and :49178 to 192.168.56.5:80, CLOSED, PID -1 **Limit:** Two HTTP-port connections to the address in the URL. Owner unknown, no time, no payload: cannot be tied to Chrome. Endpoint not contacted.
Download-folder review	2019-01-06 15:04:40	E17	shellbags: UsrClass.dat BagMRU\0\1\1\0 last write; entries Downloads and AppData **Limit:** Explorer folder view of C:\Users\IEUser\Downloads updated 10 s after Chrome started. Folder-view evidence, not a file time.
Download-folder review	14:55:45 and 14:49:40 (earlier session)	E17	shellbags: BagMRU\4 last write 14:55:45; Downloads item Modified 14:49:40 **Limit:** Earlier than this boot (15:02:47). Earlier session or clock effect; DOS stamps. Ordering hint only.
Removable-device evidence	2019-01-06 15:02:54	E12, E14, E15	USBSTOR device and instance keys, DeviceClasses and MountedDevices last write; General UDisk USB Device, F: assigned **Limit:** Device known to Windows by 15:02:54, when explorer.exe started. Not a unique drive (no serial); first and last connection times not recoverable.
Removable-device evidence	2019-01-06 15:03:07	E13	printkey MountPoints2 in the IEUser NTUSER.DAT: subkeys F and {421e7e52-…} **Limit:** IEUser shell had registered the USB volume by 15:03:07. Profile evidence, not proof of possession or identity.
Command-history evidence	2019-01-06 15:06:11	E04, E05	pslist / psscan: cmd.exe 3996 and conhost.exe 3988 created (parent explorer.exe 2524) **Limit:** Interactive command prompt opened from the IEUser shell. Start time only.
Drive mapping	2019-01-06 15:07:03	E17	shellbags: BagMRU\0 last write; F:\ at MRU position 0 **Limit:** Explorer under IEUser reached F:\ most recently at 15:07:03. Folder-view evidence; does not prove the copy took place then.
Command-history evidence	Between 15:06:11 and 15:09:06 (no per-command time)	E08, E09	cmdscan / consoles: `ipconfig`, `cd Downloads`, `copy secret_file.docx F:`; screen text `1 file(s) copied.` **Limit:** Order of commands recorded; success message recorded. No file size, content or hash; time known only within the window.
Capture boundary (context)	2019-01-06 15:07:15	E04	pslist: FTK Imager.exe 2596 started (High integrity) **Limit:** The capture tool was already running; it writes memdumpWin7.mem to the IEUser Desktop (E18) and alters memory while it runs.
Capture boundary (context)	2019-01-06 15:09:06	E02	imageinfo: image date and time **Limit:** Cut-off for the image. dllhost.exe 2228 starts at 15:09:07, one second later, showing capture smear.

Stage	UTC	Evidence	Observed fact and limit
Supported conclusion	On or before 15:09:06	E01 to E20	Chrome request for secret_file.docx from 192.168.56.5; Downloads folder; USB disk mapped as F:; copy command with success message **Limit:** Probable completed copy of secret_file.docx to F: in the IEUser session, within about 3 minutes before capture. Not proof of content, intent, removal of the drive or identity.

7. Attribution Assessment and Limitations (Question 8)

7.1 Conclusion and confidence

The evidence supports a **possible transfer attempt that probably completed**: within the IEUser session and shortly before capture, a file named secret_file.docx that Chrome had requested from 192.168.56.5 was copied from C:\Users\IEUser\Downloads to a USB flash drive mapped as F:. Confidence is **moderate** overall, with the parts as follows.

Claim	Confidence	Basis
The copy command ran in the IEUser profile and reported success	Moderate to high	cmdscan and consoles agree; screen text “1 file(s) copied.”; process tree and SID
F: was the USB flash disk	Moderate to high	USBSTOR, DeviceClasses, MountedDevices, MountPoints2 and ShellBags all point to one USB disk as F:
The file came from 192.168.56.5 through Chrome	Moderate	Strings and netscan are consistent, but the strings have no process owner and the connections have owner PID -1
The file was confidential, and its content was copied intact	Not established	No file object, hash or content in the image; the name is the only indicator
The drive left the workstation or the data was disclosed	Not established	Nothing in memory shows removal or onward use
A named person did it	Low	Only a generic profile and workstation are identified

7.2 Limitations

1. **Capture timing and capture-tool effects.** The image time (15:09:06) is only the moment the time was read while the capture ran. FTK Imager.exe (PID 2596, started 15:07:15) was running and writing the capture file to the IEUser Desktop, and dllhost.exe starts one second after the image time. Memory changed during the capture and the tool itself is part of what was captured.
2. **Clock and time-zone effects.** The earliest system processes carry a start time 3 hours later than the image time, probably from guest time synchronisation. The UUID times in the USB records show the same offset. ShellBag item stamps are DOS-format local-time values printed as UTC, and several ShellBag and Downloads times predate the boot session. Local time (UTC-8) was not used. Exact clock times for the copy therefore cannot be given; only the window 15:06:11 to 15:09:06 is supported.
3. **Profile versus human identity.** IEUser is a generic account name. There is no logon or authentication record in this evidence set, and the Console Logon SID shows only that the session was interactive. The OpenSSH service and its sshd_server profile mean a remote route to the same profile existed, although no connection to port 22 is recorded.
4. **Command success.** The success message is memory text. The file, its size and its hash are not in the image (filescan found no secret_file.docx object), and the USB drive’s contents were not examined, so the copy cannot be verified from this evidence.

5. **Network attribution.** The two connections to 192.168.56.5:80 are CLOSED, have owner PID -1, no creation time and no payload. They are consistent with the URL in the strings but cannot be tied to Chrome. No endpoint was contacted, so the nature of 192.168.56.5 is unknown.
6. **Incomplete or overwritten evidence.** The USBSTOR Properties time-stamp keys (0064, 0066, 0067) were not recoverable, so there is no first-connection or last-removal time for the drive. The MountPoints2 values are empty. cmdscan produced damaged fragments. The instance ID has no serial number, so the record identifies a generic device, not one physical drive.
7. **Strings are supplementary.** The strings search is not a Volatility plugin. The hits have no timestamp or owning process, and the `.crdownload` name shows a download began, not that it finished.
8. **Environment.** The scenario text describes Windows 10, but the image is Windows 7 SP1 x86, and the guest runs VirtualBox and Hyper-V integration services. It may be a prepared lab machine rather than a physical workstation. Findings are limited to what this image shows.

7.3 Alternative explanations

- **Authorised or routine copying.** Copying a downloaded file to a USB drive can be a normal task. Nothing in memory shows the file was confidential or that the copy was unauthorised.
- **Shared or borrowed session.** Another person could have used the IEUser session, or the session could have been left unlocked.
- **Remote access.** OpenSSH was installed and listening, so a remote user is possible, although the Console Logon SID and the interactive shell favour a local session.
- **Scripted or staged scenario.** The prepared nature of the image (earlier-session timestamps, generic profile, lab network) leaves open that the sequence was staged for training.

7.4 Recommended follow-up

- Examine the USB drive and the disk image for secret_file.docx (size, hash, file-system times) to confirm the copy and its content.
- Obtain logon, USB and Chrome artefacts from disk (Security event log, setupapi log, Chrome History and Downloads database) to date the device connection and the download and to test the shared-session explanations.
- Obtain network logs from the host serving 192.168.56.5, through authorised channels, to confirm the HTTP request without contacting it from the workstation side.

Appendix A: Labelled Screenshots

Each screenshot in this report is a terminal capture from the Kali VM, cropped to the terminal window. The caption names the plugin, profile and command; the evidence ID links it to the evidence log (Appendix D) and to the saved file in the evidence package. The original memory image is not shown or included.

Figure	Content (plugin, profile, command)	Evidence ID
Figure 1	Source file listed in the VMware shared folder (ls -l /mnt/hgfs/kali_documents/ | grep -i -E "mem|vol"): memdumpWin7.mem, 1,073,676,288 bytes.	E01
Figure 2	Acquisition block: cp -p, date -u, stat, ls -l, sha256sum, md5sum, file, chmod a-w, working-copy cp -p and hash comparison. This is the re-run; the two “Permission denied” lines show the read-only copies were not overwritten.	E01

Figure	Content (plugin, profile, command)	Evidence ID
Figure 3	acquisition_date_utc.txt: original copied 17:36:42 UTC (file birth time), hashes re-verified 17:46:30 UTC.	E01
Figure 4	Volatility 2.6.1 cloned from the Volatility Foundation repository. The import failures (Crypto.Hash, distorm3) were resolved before analysis by installing pycryptodome for Python 2.	E02
Figure 5	imageinfo, no profile: python2 vol.py -f working/memdumpWin7.mem imageinfo. Candidate profiles, KDBG, image date and time in UTC and local time.	E02
Figure 6	kdbgscan: KDBG owner-tag check True, version 15.7601; suggestions Win7SP1x86_23418, Win7SP1x86, Win7SP1x86_24000, Win7SP0x86.	E03
Figure 7	pslist, profile Win7SP1x86_23418 (saved to pslist_Win7SP1x86_23418.txt): coherent process tree.	E04
Figure 8	pslist, profile Win7SP1x86_23418, end of output (cmd.exe, conhost.exe, chrome.exe, FTK Imager.exe, dllhost.exe) and line counts for both profile runs (57 lines each).	E04
Figure 9	diff of the pslist output for Win7SP1x86_23418 and Win7SP1x86_24000 ("pslist outputs IDENTICAL") and the psscan-versus-pslist PID comparison (nothing listed).	E04, E05
Figure 10	hivelist (top) and getsids for explorer.exe (2524) and cmd.exe (3996), profile Win7SP1x86_23418.	E10, E11
Figure 11	getsids for cmd.exe (3996) and FTK Imager.exe (2596): same user SID; FTK Imager.exe runs at High mandatory level.	E11
Figure 12	psscan (end of output) and the header of netscan, profile Win7SP1x86_23418.	E05, E06
Figure 13	netscan, profile Win7SP1x86_23418: listeners, the Chrome UDP socket and the two CLOSED TCP connections to 192.168.56.5:80 with owner PID -1.	E06
Figure 14	cmdline for cmd.exe (3996), conhost.exe (3988), chrome.exe (976) and FTK Imager.exe (2596); cmdscan lines; hivelist. Profile Win7SP1x86_23418.	E07, E08, E10
Figure 15	filescan filtered for secret|.docx|IEUser, profile Win7SP1x86_23418: Desktop\memdumpWin7.mem, Links\Downloads.lnk and Chrome Download Service files; no secret_file.docx object.	E18
Figure 16	strings (ASCII and UTF-16) on the working copy filtered for secret_file. Supplementary, not a Volatility plugin: Chrome address-bar text, the URL, the .crdownload path and console text.	E19
Figure 17	consoles, profile Win7SP1x86_23418: cmd.exe (3996) history and screen buffer: ipconfig, cd Downloads, copy secret_file.docx F: and "1 file(s) copied."	E09
Figure 18	printkey MountedDevices (SYSTEM hive 0x87c1c008): key last updated 2019-01-06 15:02:54 UTC.	E12
Figure 19	printkey MountedDevices: \DosDevices\F: decodes to USBSTOR#Disk&Ven_General&Prod_UDisk.	E12
Figure 20	printkey MountPoints2 in the IEUser NTUSER.DAT (0x8b5d39b0): last updated 15:03:07 UTC; subkeys F and {421e7e52-…}.	E13
Figure 21	printkey Select (Current = 1, so ControlSet001 is live) and USBSTOR with its device-class key.	E14, E16
Figure 22	printkey on the USBSTOR instance key 6&1bec0f48&0&_&0: FriendlyName, ContainerID, HardwareID.	E14
Figure 23	printkey on the instance Properties key and on DeviceClasses {53f56307-…}: last updated 15:02:54 UTC.	E14, E15
Figure 24	shellbags: BagMRU\0 in the IEUser UsrClass.dat lists volumes C:, A:, D:, E:, F:, with F:\ at MRU position 0 (key last updated 15:07:03 UTC).	E17
Figure 25	shellbags: Downloads bags (BagMRU\4 last updated 14:55:45 UTC; BagMRU\0\1\1\0 last updated 15:04:40 UTC) and the folder's own modified, created and accessed stamps.	E17
Figure 26	Saved shellbags text searched for F:, Downloads and related terms (lines 39, 57, 184); the three "requested key could not be found" results above it are the Properties time-stamp keys (0064, 0066, 0067), which were not recoverable.	E17, E14
Figure 27	Evidence records listed, hashed into a manifest and zipped for review (58 KB; no memory image).	E20

Appendix B: Selected Volatility Output

Extracts from the saved files in `evidence_records/` (profile Win7SP1x86_23418). Long lines are shortened with "…"; the files are complete in the evidence package.

B1. imageinfo (imageinfo.txt)

```
Volatility Foundation Volatility Framework 2.6.1
INFO    : volatility.debug    : Determining profile based on KDBG search...
```

```
          Suggested Profile(s) : Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86
                     AS Layer1 : IA32PagedMemoryPae (Kernel AS)
                     AS Layer2 : FileAddressSpace
(/home/kali/SBTDF204_CS3/working/memdumpWin7.mem)
                      PAE type : PAE
                           DTB : 0x185000L
                          KDBG : 0x82972c68L
          Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0x82973d00L
             KUSER_SHARED_DATA : 0xffdf0000L
           Image date and time : 2019-01-06 15:09:06 UTC+0000
     Image local date and time : 2019-01-06 07:09:06 -0800
```

B2. cmdscan, conhost.exe 3988 (cmdscan_Win7SP1x86_23418.txt)

```
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????????
Cmd #37 @ 0x10cff0: ???????
```

B3. consoles, cmd.exe 3996, end of screen buffer (consoles_Win7SP1x86_23418.txt)

```
Tunnel adapter isatap.{7FD51246-C28D-46B6-AA98-EB2C150C388B}:
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
Tunnel adapter Teredo Tunneling Pseudo-Interface:
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
C:\Users\IEUser>cd Downloads
C:\Users\IEUser\Downloads>copy secret_file.docx F:
        1 file(s) copied.
C:\Users\IEUser\Downloads>
```

B4. netscan: connections to 192.168.56.5 and the Chrome socket (netscan_Win7SP1x86_23418.txt)

```
Offset(P)          Proto    Local Address                  Foreign Address      State
Pi…
0x3e081f50         UDPv4    192.168.56.8:56647             *:*
2388     chrome.exe …
0x3e043b10         TCPv4    192.168.56.8:49177             192.168.56.5:80      CLOSED
-1
0x3ee1aa88         TCPv4    0.0.0.0:22                     0.0.0.0:0            LISTENING
2016     sshd.exe
0x3ee1bc18         TCPv4    0.0.0.0:22                     0.0.0.0:0            LISTENING
2016     sshd.exe
0x3ee1bc18         TCPv6    :::22                          :::0                 LISTENING
2016     sshd.exe
0x3fcba008         TCPv4    192.168.56.8:49178             192.168.56.5:80      CLOSED
-1
```

B5. getsids for cmd.exe 3996 (getsids_Win7SP1x86_23418.txt)

```
cmd.exe (3996): S-1-5-21-1716914095-909560446-1177810406-1000 (IEUser)
cmd.exe (3996): S-1-5-21-1716914095-909560446-1177810406-513 (Domain Users)
cmd.exe (3996): S-1-5-32-544 (Administrators)
cmd.exe (3996): S-1-2-1 (Console Logon (Users who are logged onto the physical console))
cmd.exe (3996): S-1-16-8192 (Medium Mandatory Level)
```

B6. USBSTOR instance key (printkey_USBSTOR_instance_Win7SP1x86_23418.txt)

```
---------------------------
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: 6&1bec0f48&0&_&0 (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) Device Parameters
  (S) LogConf
  (S) Properties
  (V) Control

Values:
REG_SZ        DeviceDesc      : (S) @disk.inf,%disk_devdesc%;Disk drive?
REG_DWORD     Capabilities    : (S) 0
REG_MULTI_SZ  HardwareID      : (S) ['USBSTOR\\DiskGeneral_UDisk_____________5.00',
'USBSTOR\\DiskGeneral_UDisk_________…
REG_MULTI_SZ  CompatibleIDs   : (S) ['USBSTOR\\Disk', 'USBSTOR\\RAW', '', '']
REG_SZ        ContainerID     : (S) {421e7e37-11dd-11e9-b3cf-806e6f6e6963}?
REG_DWORD     ConfigFlags     : (S) 0
REG_SZ        ClassGUID       : (S) {4d36e967-e325-11ce-bfc1-08002be10318}?
REG_SZ        Class           : (S) DiskDrive?
REG_SZ        Mfg             : (S) @disk.inf,%genmanufacturer%;(Standard disk drives)?
```

B7. DeviceClasses and MountPoints2

```
Last updated: 2019-01-06 15:02:54 UTC+0000
  (S) ##?#USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0#{53f56307-b6bf-
11d0-94f2-00a0c91efb8b}

---------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
  (S) CPC
  (S) F
  (S) {421e7e52-11dd-11e9-b3cf-08002709e15d}
  (S) {762f4ebc-60ea-11e5-83af-806e6f6e6963}
  (S) {8358fe6d-60fa-11e5-bb4a-806e6f6e6963}
  (S) {a5b8a980-608c-11e5-a266-806e6f6e6963}
  (S) {a5b8a983-608c-11e5-a266-806e6f6e6963}

Values:
```

B8. shellbags: BagMRU\0 and the Downloads bags (shellbags_Win7SP1x86_23418.txt)

```
Registry: \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0
Last updated: 2019-01-06 15:07:03 UTC+0000
Value   Mru   Entry Type     Path
------- ----- -------------- ----
1       1     Volume Name    C:\
0       4     Volume Name    A:\
3       3     Volume Name    D:\
2       2     Volume Name    E:\
4       0     Volume Name    F:\
**************************************************************************
...
Registry: \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\4
Last updated: 2019-01-06 14:55:45 UTC+0000
Value   Mru   Entry Type     GUID                                   GUID Description
Folder IDs
------- ----- -------------- -------------------------------------- --------------------
----------
0       0     Folder         374de290-123f-4565-9164-39c4925e467b     Downloads
EXPLORER
**************************************************************************
...
```

```
Registry: \??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat
Key: Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1\0
Last updated: 2019-01-06 15:04:40 UTC+0000
Value   Mru   File Name      Modified Date                  Create Date
Access Date                    File Attr                P…
------- ----- -------------- ------------------------------ ------------------------------
------------------------------ ------------------------- -…
1       0     DOWNLO~1       2019-01-06 14:49:40 UTC+0000   2015-09-21 09:17:34 UTC+0000
2019-01-06 14:49:40 UTC+0000   RO, DIR                   C…
0       1     AppData        2015-09-21 09:17:34 UTC+0000   2015-09-21 09:17:34 UTC+0000
2015-09-21 09:17:34 UTC+0000   HID, NI, DIR              C…
*****************************************************************************
```

B9. strings, selected hits (strings_utf16_secret_file.txt, strings_ascii_secret_file.txt)

```
/secret_file.docxhttps://www.google.com/search?q=192.168.56.5%2Fsecret_file.docx&oq=192.168
.56.5%2Fsecret_file.docx&a…
ttp://192.168.56.5/secret_file.docx
http://192.168.56.5/secret_file.docx
/secret_file.docxhttps://www.google.com/search?q=192.168.56.5%2Fsecret_file.docx&oq=192.168
.56.5%2Fsecret_file.docx&a…
secret_file.docx.crdownloadexe-1177810406-1000}
ers\IEUser\Downloads\secret_file.docx.crdownload\
secret_file.docx.crdownload68fd06275.tmp
file:C:/Users/IEUser/Downloads/secret_file.docx
C:\Users\IEUser\Downloads\secret_file.docx.crdownload
C:\Users\IEUser\Downloads\secret_file.docx
secret_file.docx.crdownload68fd06275.tmp
file:C:/Users/IEUser/Downloads/secret_file.docx
F:\secret_file.docx
secret_file.docx.crdownloadexe-1177810406-1000}
```

Appendix C: Completed Timeline Table (UTC)

Stage, UTC timestamp, evidence ID, source artefact, observed fact, and interpretation and limit, as required by the case study template. Times are as printed by Volatility. “Undated” rows have no time in the evidence.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Browser or process evidence	2019-01-06 15:04:30	E04, E05	pslist / psscan	pslist / psscan: chrome.exe 2388 (parent explorer.exe 2524); children 2380, 1280 (15:04:30), 2912 (15:04:32), renderer 976 (15:06:23)	Chrome process tree created from the IEUser shell 4 min 36 s before image time. Process creation only: no URL, page or download is shown.
Browser or process evidence	Undated (RAM strings)	E19	strings -a, strings -a -e l	strings: Chrome address-bar text `http://192.168.56.5/secret_file.docx` and prefixes of it; `C:\Users\IEUser\Downloads\secret_file.docx.crdownload`; `F:\secret_file.docx`	Consistent with Chrome requesting the file and starting a download. No timestamp, no process owner (not Volatility output). `.crdownload` shows a download began, not that it finished.
Browser or process evidence	Undated (closed)	E06	netscan	netscan 0x3e043b10 and 0x3fcba008: TCPv4 192.168.56.8:49177 and :49178 to 192.168.56.5:80, CLOSED, PID -1	Two HTTP-port connections to the address in the URL. Owner unknown, no time, no payload: cannot be tied to Chrome. Endpoint not contacted.
Download-folder review	2019-01-06 15:04:40	E17	shellbags (BagMRU\0\1\1\0)	shellbags: UsrClass.dat BagMRU\0\1\1\0 last write; entries Downloads and AppData	Explorer folder view of C:\Users\IEUser\Downloads updated 10 s after Chrome started. Folder-view evidence, not a file time.
Download-folder review	14:55:45 and 14:49:40 (earlier session)	E17	shellbags (BagMRU\4; Downloads item)	shellbags: BagMRU\4 last write 14:55:45; Downloads item Modified 14:49:40	Earlier than this boot (15:02:47). Earlier session or clock effect; DOS stamps. Ordering hint only.
Removable-device evidence	2019-01-06 15:02:54	E12, E14, E15	printkey: USBSTOR, DeviceClasses, MountedDevices	USBSTOR device and instance keys, DeviceClasses and MountedDevices last write; General UDisk USB Device, F: assigned	Device known to Windows by 15:02:54, when explorer.exe started. Not a unique drive (no serial); first and last connection times not recoverable.
Removable-device evidence	2019-01-06 15:03:07	E13	printkey: MountPoints2 (IEUser NTUSER.DAT)	printkey MountPoints2 in the IEUser NTUSER.DAT: subkeys F and {421e7e52-…}	IEUser shell had registered the USB volume by 15:03:07. Profile evidence, not proof of possession or identity.
Command-history evidence	2019-01-06 15:06:11	E04, E05	pslist / psscan	pslist / psscan: cmd.exe 3996 and conhost.exe 3988 created (parent explorer.exe 2524)	Interactive command prompt opened from the IEUser shell. Start time only.
Drive mapping	2019-01-06 15:07:03	E17	shellbags (BagMRU\0)	shellbags: BagMRU\0 last write; F:\ at MRU position 0	Explorer under IEUser reached F:\ most recently at 15:07:03. Folder-view evidence; does not prove the copy took place then.
Command-history evidence	Between 15:06:11 and 15:09:06 (no per-command time)	E08, E09	cmdscan; consoles	cmdscan / consoles: `ipconfig`, `cd Downloads`, `copy secret_file.docx F:`; screen text `1 file(s) copied.`	Order of commands recorded; success message recorded. No file size, content or hash; time known only within the window.
Capture boundary (context)	2019-01-06 15:07:15	E04	pslist	pslist: FTK Imager.exe 2596 started (High integrity)	The capture tool was already running; it writes memdumpWin7.mem to the IEUser Desktop (E18) and alters memory while it runs.
Capture boundary (context)	2019-01-06 15:09:06	E02	imageinfo	imageinfo: image date and time	Cut-off for the image. dllhost.exe 2228 starts at 15:09:07, one second later, showing capture smear.
Supported conclusion	On or before 15:09:06	E01 to E20	All of the above	Chrome request for secret_file.docx from 192.168.56.5; Downloads folder; USB disk mapped as F:; copy command with success message	Probable completed copy of secret_file.docx to F: in the IEUser session, within about 3 minutes before capture. Not proof of content, intent, removal of the drive or identity.

Appendix D: Evidence Log

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E01	cp -p, date -u, stat, ls -l, sha256sum, md5sum, file, chmod	original/ and working/memdumpWin7.mem; evidence_records/ acquisition_date_utc, file_stat, file_size, file_type, file_perms_after_chmod, md5_original, sha256_original, sha256_working	Size 1,073,676,288 bytes; SHA-256 500bc1ba…099495e; MD5 ac7eaa46…; copied 17:36:42 UTC; originals read-only; hashes match	Figures 1 to 3; Section 3
E02	Volatility 2.6.1: imageinfo	imageinfo.txt; volatility_version.txt	Win7 SP1 x86 PAE; candidates Win7SP1x86_23418, Win7SP0x86, Win7SP1x86_24000, Win7SP1x86; image time 2019-01-06 15:09:06 UTC (local 07:09:06 -0800)	Figures 4, 5; Appendix B1
E03	kdbgscan	kdbgscan.txt	KDBG 0x2972c68, owner-tag check True, version 15.7601; four profile suggestions; stale KDBG copies at 0xd6c7c68 and 0xec94c68	Figure 6
E04	pslist (Win7SP1x86_23418 and Win7SP1x86_24000)	pslist_Win7SP1x86_23418.txt; pslist_Win7SP1x86_24000.txt	57 lines each, identical; chrome.exe 2388 (15:04:30), cmd.exe 3996 and conhost.exe 3988 (15:06:11), FTK Imager.exe 2596 (15:07:15), dllhost.exe 2228 (15:09:07); early-boot processes 18:02:44 to 18:02:45	Figures 7 to 9
E05	psscan; PID comparison with pslist	psscan_Win7SP1x86_23418.txt	Same PIDs as pslist; only terminated process cygrunsrv.exe 1984 (exit 15:02:49); no hidden process	Figures 9, 12
E06	netscan	netscan_Win7SP1x86_23418.txt	No ESTABLISHED connection; chrome.exe 2388 UDP 192.168.56.8:56647 (15:08:32); two CLOSED TCPv4 192.168.56.8:49177 and :49178 to 192.168.56.5:80, PID -1; sshd.exe 2016 on port 22	Figures 12, 13; Appendix B4
E07	cmdline	cmdline_Win7SP1x86_23418.txt	cmd.exe 3996: C:\Windows\system32\cmd.exe; conhost 3988; chrome.exe 976 renderer; FTK Imager.exe 2596 from C:\Program Files\AccessData\FTK Imager\	Figure 14
E08	cmdscan	cmdscan_Win7SP1x86_23418.txt	conhost 3988 / cmd.exe: Cmd #0 ipconfig, #1 cd Downloads, #2 copy secret_file.docx F:; damaged fragments in other slots	Figure 14; Appendix B2
E09	consoles	consoles_Win7SP1x86_23418.txt	Screen buffer: C:\Users\IEUser>ipconfig (192.168.56.8), cd Downloads, copy secret_file.docx F:, “1 file(s) copied.”	Figure 17; Appendix B3
E10	hivelist	hivelist_Win7SP1x86_23418.txt	SYSTEM 0x87c1c008; IEUser ntuser.dat 0x8b5d39b0 and UsrClass.dat 0xa620b008; sshd_server ntuser.dat 0xa65269c8 and UsrClass.dat 0xa6539260	Figures 10, 14
E11	getsids -p 3996,2524,2596	getsids_Win7SP1x86_23418.txt	IEUser SID S-1-5-21-1716914095-909560446-1177810406-1000; Administrators, Console Logon; FTK Imager.exe at High integrity	Figures 10, 11; Appendix B5
E12	printkey -o 0x87c1c008 -K MountedDevices	printkey_MountedDevices_Win7SP1x86_23418.txt	F: maps to USBSTOR#Disk&Ven_General&Prod_UDisk&Rev_5.00#6&1bec0f48&0&_&0; volume {421e7e52-…}; last write 15:02:54	Figures 18, 19
E13	printkey -o 0x8b5d39b0 -K …\Explorer\MountPoints2	printkey_MountPoints2_IEUser_Win7SP1x86_23418.txt	Subkeys CPC, F, {421e7e52-…} and four older GUIDs; last write 15:03:07	Figure 20; Appendix B7
E14	printkey -K ControlSet001\Enum\USBSTOR (device, instance, Properties)	printkey_USBSTOR_*.txt, printkey_USBSTOR_UDisk_*.txt, printkey_USBSTOR_instance_*.txt, printkey_USBSTOR_timestamps_*.txt	General UDisk USB Device; ContainerID {421e7e37-…}; keys last write 15:02:54; Properties time-stamp keys (0064, 0066, 0067) not found	Figures 21 to 23, 26; Appendix B6

Evidence ID	Tool and plugin	Source record or path	Observed fact	Screenshot or appendix reference
E15	printkey -K ControlSet001\Control\DeviceClasses\{53f56307-…}	printkey_DeviceClasses_disk_Win7SP1x86_23418.txt	USBSTOR interface entry for the same device; last write 15:02:54	Figure 23; Appendix B7
E16	printkey -K Select	printkey_Select_Win7SP1x86_23418.txt	Current 1, Default 1, Failed 0, LastKnownGood 2: ControlSet001 is live	Figure 21
E17	shellbags (text and HTML)	shellbags_Win7SP1x86_23418.txt; shellbags_Win7SP1x86_23418.html	BagMRU\0: volumes with F:\ at MRU 0 (15:07:03); Downloads bags (15:04:40, 14:55:45); Downloads item Modified 14:49:40, Created 2015-09-21 09:17:34	Figures 24 to 26; Appendix B8
E18	filescan	filescan_Win7SP1x86_23418.txt; filescan_downloads_secret_Win7SP1x86_23418.txt (empty)	No “secret” or “.docx” object; Desktop\memdumpWin7.mem; Downloads folder and Downloads.lnk; root of \Device\HarddiskVolume2\	Figure 15
E19	strings -a; strings -a -e l (supplementary; not Volatility)	strings_ascii_secret_file.txt; strings_utf16_secret_file.txt	http://192.168.56.5/secret_file.docx and prefixes; Downloads\secret_file.docx.crdownload; F:\secret_file.docx	Figure 16; Appendix B9
E20	sha256sum manifest; zip	evidence_manifest_sha256.txt; CS3_records_for_review.zip	SHA-256 of every output file listed above	Figure 27

Appendix E: Command Record

Commands as run in the Kali VM (shell variables: `IMG=~/SBTDF204_CS3/working/memdumpWin7.mem`, `EV=~/SBTDF204_CS3/evidence_records`, `P=Win7SP1x86_23418`).

Acquisition and preservation

```
vmware-hgfsclient; sudo mkdir -p /mnt/hgfs
sudo vmhgfs-fuse .host:/ /mnt/hgfs -o allow_other -o uid=$(id -u) -o gid=$(id -g)
cd ~/SBTDF204_CS3
cp -p /mnt/hgfs/kali_documents/memdumpWin7.mem original/memdumpWin7.mem
date -u | tee evidence_records/acquisition_date_utc.txt
stat original/memdumpWin7.mem | tee evidence_records/file_stat.txt
ls -l original/memdumpWin7.mem | tee evidence_records/file_size.txt
sha256sum original/memdumpWin7.mem | tee evidence_records/sha256_original.txt
md5sum original/memdumpWin7.mem | tee evidence_records/md5_original.txt
file original/memdumpWin7.mem | tee evidence_records/file_type.txt
chmod a-w original/memdumpWin7.mem
ls -l original/memdumpWin7.mem | tee evidence_records/file_perms_after_chmod.txt
cp -p original/memdumpWin7.mem working/memdumpWin7.mem
sha256sum working/memdumpWin7.mem | tee evidence_records/sha256_working.txt
diff <(cut -d' ' -f1 evidence_records/sha256_original.txt) <(cut -d' ' -f1
evidence_records/sha256_working.txt) && echo "HASHES MATCH"
{ echo "Original copied into case folder (file birth time, UTC):"; date -u -d "$(stat -c %w
original/memdumpWin7.mem)"; echo "Hashes re-verified (UTC):"; date -u; } | tee
evidence_records/acquisition_date_utc.txt
```

Volatility 2 setup

```
git clone https://github.com/volatilityfoundation/volatility.git volatility2   # Volatility
Foundation Volatility Framework 2.6.1
cd ~/volatility2; python2 vol.py --info | head -20
python2 -m pip install --user pycryptodome   # for shellbags, getsids and related plugins
(pip for Python 2 installed with get-pip.py --user)
```

Volatility plugins (each piped through grep -v "Failed to import" and tee)

```
cd ~/volatility2
python2 vol.py -f $IMG imageinfo 2>&1 | grep -v "Failed to import" | tee $EV/imageinfo.txt
python2 vol.py -f $IMG kdbgscan 2>&1 | grep -v "Failed to import" | tee $EV/kdbgscan.txt
python2 vol.py -f $IMG --profile=Win7SP1x86_23418 pslist  ... | tee
$EV/pslist_Win7SP1x86_23418.txt
python2 vol.py -f $IMG --profile=Win7SP1x86_24000 pslist  ... >
$EV/pslist_Win7SP1x86_24000.txt
diff $EV/pslist_Win7SP1x86_23418.txt $EV/pslist_Win7SP1x86_24000.txt && echo "pslist
outputs IDENTICAL"
python2 vol.py -f $IMG --profile=$P psscan   2>&1 | grep -v "Failed to import" | tee
$EV/psscan_$P.txt
comm -13 <(awk '$3 ~ /^[0-9]+$/ {print $3}' $EV/pslist_$P.txt | sort -u) <(awk '$3 ~ /^[0-
9]+$/ {print $3}' $EV/psscan_$P.txt | sort -u)
python2 vol.py -f $IMG --profile=$P netscan  ... | tee $EV/netscan_$P.txt
python2 vol.py -f $IMG --profile=$P cmdline  ... | tee $EV/cmdline_$P.txt
python2 vol.py -f $IMG --profile=$P cmdscan  ... | tee $EV/cmdscan_$P.txt
python2 vol.py -f $IMG --profile=$P consoles ... | tee $EV/consoles_$P.txt
python2 vol.py -f $IMG --profile=$P hivelist ... | tee $EV/hivelist_$P.txt
python2 vol.py -f $IMG --profile=$P getsids -p 3996,2524,2596 ... | tee $EV/getsids_$P.txt
python2 vol.py -f $IMG --profile=$P filescan ... > $EV/filescan_$P.txt
python2 vol.py -f $IMG --profile=$P printkey -o 0x87c1c008 -K 'Select' ...
python2 vol.py -f $IMG --profile=$P printkey -o 0x87c1c008 -K 'ControlSet001\Enum\USBSTOR'
...
python2 vol.py -f $IMG --profile=$P printkey -o 0x87c1c008 -K
'ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00' ...
INST='ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_&0'
python2 vol.py -f $IMG --profile=$P printkey -o 0x87c1c008 -K "$INST" 2>&1 | grep -a -v
"Failed to import" | tee $EV/printkey_USBSTOR_instance_$P.txt
python2 vol.py -f $IMG --profile=$P printkey -o 0x87c1c008 -K 'MountedDevices' ...
python2 vol.py -f $IMG --profile=$P printkey -o 0x87c1c008 -K
'ControlSet001\Control\DeviceClasses\{53f56307-b6bf-11d0-94f2-00a0c91efb8b}' ...
python2 vol.py -f $IMG --profile=$P printkey -o 0x8b5d39b0 -K
'Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2' ...
```

```
python2 vol.py -f $IMG --profile=$P shellbags 2>&1 | grep -v "Failed to import" | tee
$EV/shellbags_$P.txt
python2 vol.py -f $IMG --profile=$P shellbags --output=html --output-
file=$EV/shellbags_$P.html
```

Supplementary strings search and packaging

```
strings -a $IMG | grep -i "secret_file" | head -50 > $EV/strings_ascii_secret_file.txt
strings -a -e l $IMG | grep -i "secret_file" | head -50 > $EV/strings_utf16_secret_file.txt
cd $EV && sha256sum * > evidence_manifest_sha256.txt
cd ~ && zip -r CS3_records_for_review.zip SBTDF204_CS3/evidence_records
```

The three `printkey` commands for the Properties time-stamp keys (0064, 0066, 0067) returned “The requested key could not be found in the hive(s) searched” and are recorded in `printkey_USBSTOR_timestamps_Win7SP1x86_23418.txt`.

Appendix F: Evidence Package and Handling Notes

- The evidence holds this report (PDF), the cropped screenshots used as figures, every Volatility and strings output file in `evidence_records/` (including the shellbags HTML report and the SHA-256 manifest. It does **not** contain the original or working memory image, as the brief does not authorise uploading it.
- Screenshots were cropped to the terminal window only; the Windows taskbar, the VMware library pane and unrelated desktop items were removed. The IP addresses, user and profile names, SIDs and file names shown are part of the case evidence. The Chrome address-bar text in the strings file includes an unrelated browser API key that ships with Chrome; it is kept in the raw strings file so the output is unaltered, and is not reproduced in this report.
- No live service was contacted: no IP address, host, account, share or person named in the output was contacted, and no recovered file was executed. No password or hash-cracking step was used.
- Work was done individually, in the Kali lab VM, offline from the evidence.

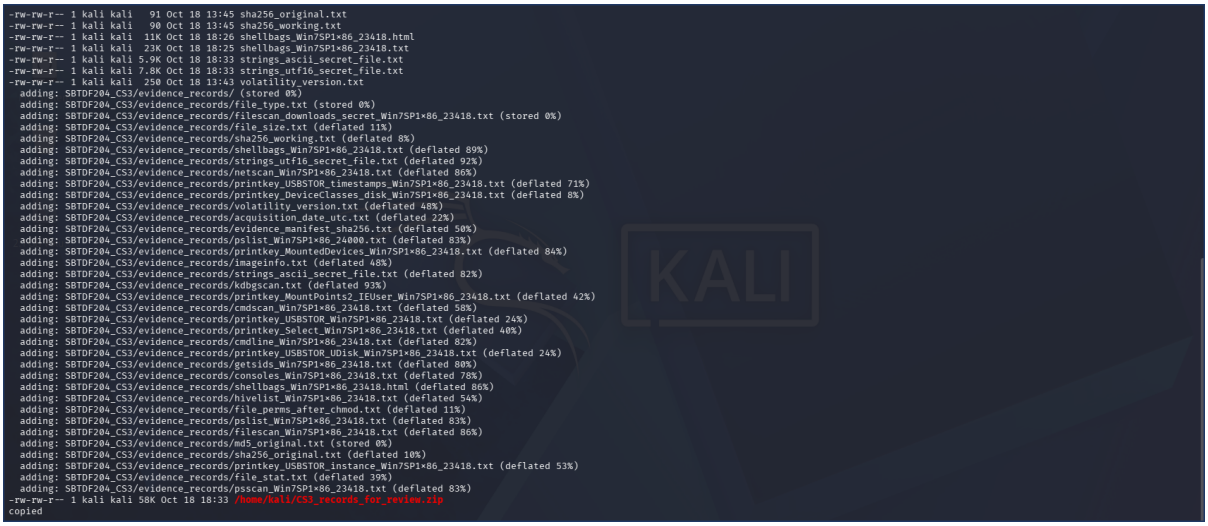


Figure 27: Evidence records listed, hashed into a manifest and zipped for review (58 KB; no memory image). (E20)